Modeling Dynamic Systems

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1 Dynamics of mechanical systems (Franklin *et al.* section 2.1 (7th ed.))

- Systems with storage introduce differential equations:

$$\frac{d\,\text{Stored Quantity}}{dt} = \text{Force or Rate of Flow}$$

Examples (stored quantity is energy)

- $E = \frac{1}{2} C v^2 \longrightarrow$ Current = Capacitance $* \frac{d}{dt}$ Voltage , $\frac{dE}{dt} = v(t) \times i(t)$
- $E = \frac{1}{2} M v^2 \longrightarrow$ Force = Mass $* \frac{d}{dt}$ Velocity , $\frac{dE}{dt} = v(t) \times f(t)$

  Storage is often energy storage, but also business systems (inventory), economics (accounts), etc.

Why does the Controls Engineer need dynamics ?

- A model of the plant to be controlled is needed for design control.

How ?

- From the physics governing the plant, such as:
  - From (for linear motion)

$$F(t) = mA(t) \quad \left(F(t) = m \frac{d^2 x(t)}{dt^2} \right) \tag{1}$$

  - Or (for rotary motion)

$$M(t) = I\,\ddot{\Theta}(t) \tag{2}$$

  - And similar relations.

Signal	Description	Units	Signal	Description	Units
$v(t)$	voltage	[volts]	$i(t)$	current	[amps]
$x(t)$	position	[meters]	$F(t)$	force	[Newtons]
$\dot{x}(t) = v(t)$	velocity	[m/sec]			
$\theta(t)$	angle	[radians]	$T(t)$ or $M(t)$	torque (or moment)	[Newton-meters]
$\dot{\theta}(t) = \omega(t)$	angular velocity	[rad/sec]			

Table 1: Some signals used to analyze motion control systems.

Linear motion			Rotational motion		
Term	Modeling equation	Units	Term	Modeling equation	Units
Mass	$m\ddot{x}(t) = f_{net}(t)$	$[Kg] = \left[\frac{N}{m/s^2}\right] = \left[\frac{N-s^2}{m}\right]$	Inertia	$I\ddot{\theta}(t) = M_{net}(t)$	$\left[Kg-m^2\right] = \left[\frac{N-m}{rad/sec^2}\right] = \left[\frac{N-m-s^2}{rad}\right]$
Spring	$f_k(t) = k(x_1(t) - x_2(t))$	$\left[\frac{N}{m}\right]$	Rotational spring	$M_k(t) = k(\theta_1(t) - \theta_2(t))$	$\left[\frac{N-m}{rad}\right]$
Damper	$f_b(t) = b(\dot{x}_1(t) - \dot{x}_2(t))$	$\left[\frac{N}{m/s}\right] = \left[\frac{N-s}{m}\right]$	Rotational damper	$M_b(t) = b\left(\dot{\theta}_1(t) - \dot{\theta}_2(t)\right)$	$\left[\frac{N-m}{rad/sec}\right] = \left[\frac{N-m-s}{rad}\right]$

Table 2: Basic parameters of mechanical systems.

1.1 Example 2.1 (7th ed.), Car cruise control

Schematic:

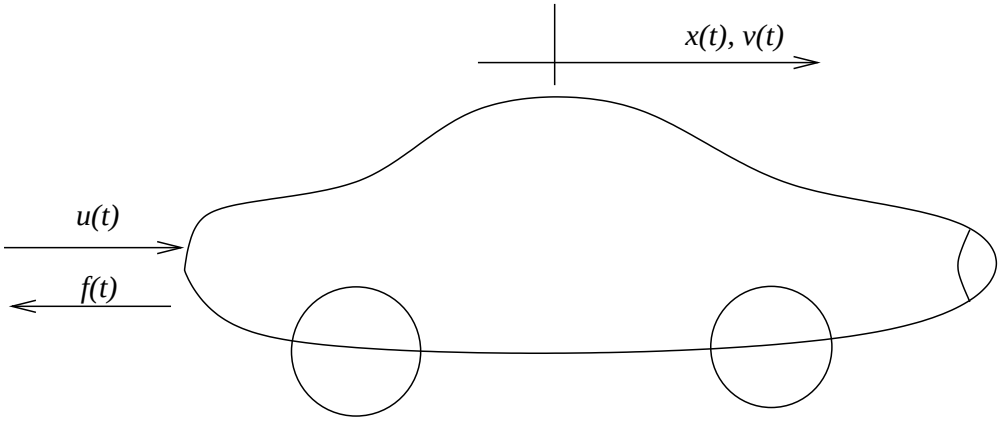


Figure 1: Schematic diagram for automobile cruise control.

Signals:

Signal	Units	Description
$u(t)$	[N]	Applied Force
$v(t)$	[m/s]	Car Velocity
$f(t)$	[N]	Drag Force
$F_{net}(t)$	[N]	Net force on Car

Table 3: Cruise control signals.

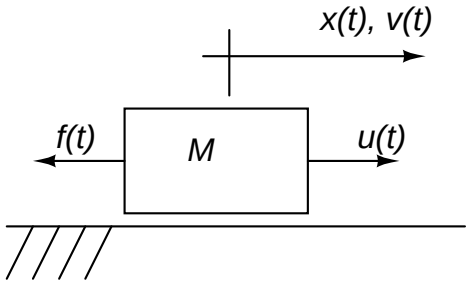
Parameters:

Parameter	Units	Description
M	[Kg]	Mass
b	$\left[\frac{N}{m/s}\right]$, or $\left[\frac{N-s}{m}\right]$	Drag Coefficient

Table 4: Cruise control parameters.

1.1.1 Free-body diagram and deriving the Governing Equation

- Draw the free-body diagram to list each force acting on a mass.



Elements
M: Mass of car
x(t),v(t): position and velocity
u(t): Traction force generated by motor
f(t): External force on car (friction or drag)

Figure 2: Free-body diagram.

- Net force is the vector sum of applied forces

$$u(t) - f(t) = F_{net}(t) \tag{3}$$

- Newton's second Law (acceleration given by force over mass)

$$F_{net}(t) = MA(t) = M\ddot{x}(t) = u(t) - f(t) \tag{4}$$

- Constituent relation for a damper (friction or drag element):

$$f(t) = b\,v(t) = b\,\dot{x}(t) \tag{5}$$

- Combine equations and put ***Governing Equation*** in standard form:

$$M\ddot{x}(t) + b\dot{x}(t) = u(t) \tag{6}$$

Dependent variables on left (Output variable) | Independent variables on right (Input variables)

- Using $v(t) = \dot{x}(t)$, Eqn (6) can also be written:

$$\dot{v}(t) + \frac{b}{M}v(t) = \frac{1}{M}u(t) \tag{7}$$

2 Writing the transfer function (Franklin *et al.* section 2.1 (7th ed.))

- The transfer function utilizes the ***Laplace Transform*** to capture the differential equation in a simpler form.
- We'll explore the Laplace transform and transfer function more fully in chapter 3, here is a basic introduction.

2.1 Deriving an example transfer function, car cruise control

- Starting with the governing equation for the system:

$$\dot{v}(t) + \frac{b}{M} v(t) = \frac{1}{M} u(t) \qquad (7, \text{ repeated})$$

  - Each of the signals has a Laplace transform:

$$V(s) = \mathcal{L}\{v(t)\} \tag{8}$$
$$U(s) = \mathcal{L}\{u(t)\} \tag{9}$$

  - We will also use the derivative theorem of the Laplace transform (LT):

$$\mathcal{L}\{\dot{v}(t)\} = sV(s) \tag{10}$$

  - Taking the LT of both sides of Eqn (7)

$$\mathcal{L}\left\{\dot{v}(t) + \frac{b}{M} v(t)\right\} = \mathcal{L}\left\{\frac{1}{M} u(t)\right\} \tag{11}$$

  - Using the linearity of the LT, and the derivative theorem, we derive a relationship between the LT of the input and the LT of the output

$$sV(s) + \frac{b}{M} V(s) = \frac{1}{M} U(s) \; . \tag{12}$$

- Starting with

$$sV(s)+\frac{b}{M}V(s)=\frac{1}{M}U(s) \qquad (12, \text{ repeated})$$

Group the terms on the left and the right, pulling out constants and powers of s

$$\left(s+\frac{b}{M}\right)V(s)=\frac{1}{M}U(s) \qquad (13)$$

- The transfer function is the ratio of the output LT over the input LT:

$$\frac{\text{Output LT}}{\text{Input LT}} \;:\; \frac{V(s)}{U(s)}=G_{uv}(s)=\frac{1/M}{s+b/M} \qquad (14)$$

2.2 A transfer function can be between any input and any output signal

- Going back to example 2.1 (automobile cruise control, figure 1), the position may be the output of interest, giving us a 2nd order differential equation

$$\ddot{x}(t)+\frac{b}{M}\dot{x}(t)=\frac{1}{M}u(t) \quad \overset{\mathcal{L}}{\longleftrightarrow} \quad s^2X(s)+\frac{b}{M}sX(s)=\frac{1}{M}U(s)$$

- Group the terms on the left and the right

$$\left(s^2+\frac{b}{M}s\right)X(s)=\frac{1}{M}U(s) \qquad (15)$$

which gives the transfer function:

$$\frac{\text{Output}}{\text{Input}}=\frac{X(s)}{U(s)}=G_{ux}(s)=\frac{1/M}{s^2+(b/M)\,s}=\frac{1/M}{s\,(s+b/M)} \qquad (16)$$

2.3 Double check by checking units of the transfer function

Units must balance in ***all*** equations modeling physical systems.

- The same is true of transfer functions (use tables 3 and 4 for units of mechanical parameters);
- The units of the Laplace variable s are $\left[\text{sec}^{-1}\right]$.
- For the TF of Eqn (16) the units are given by:

$$G_{ux}(s) = \frac{1/M}{s\,(s+b/M)} \;:\; \frac{\text{Kg}^{-1}}{\text{s}^{-1}\left(\text{s}^{-1}+\frac{\left[\frac{\text{N-s}}{\text{m}}\right]}{\text{Kg}}\right)} \tag{17}$$

Note: the units of s are s^{-1} . Perhaps another variable should have been chosen by Laplace to be the Laplace variable; but he chose s. Controls is full of notation conflicts.

The units of Eqn (17) must be reduced. Using $F = MA$, we can write:

$$1\,[\text{N}] = 1\,[\text{Kg}]\left[\frac{\text{m}}{\text{s}^2}\right] = 1\left[\frac{\text{Kg}-\text{m}}{\text{s}^2}\right] \tag{18}$$

Rearranging gives:

$$1\,[\text{Kg}] = 1\left[\frac{\text{N}-\text{s}^2}{\text{m}}\right]. \tag{19}$$

So the units of Eqn (17) become:

$$\begin{aligned}\left[\frac{\text{Kg}^{-1}}{\text{s}^{-1}\left(\text{s}^{-1}+\frac{\left[\frac{\text{N-s}}{\text{m}}\right]}{\text{Kg}}\right)}\right] &= \left[\frac{\left[\frac{m}{N-s^2}\right]}{s^{-1}\left(s^{-1}+\frac{\left[\frac{N-s}{m}\right]}{\left[\frac{\text{N-s}^2}{\text{m}}\right]}\right)}\right] \\ &= \left[\frac{\left[\frac{m}{N-s^2}\right]}{s^{-1}\left(s^{-1}+s^{-1}\right)}\right] = \left[\frac{\left[\frac{m}{N-s^2}\right]}{s^{-2}}\right] = \left[\frac{\text{m}}{\text{N}}\right]\end{aligned} \tag{20}$$

- The units on transfer function are given as

$$G_{ux}(s) = \frac{1/M}{s\,(s+b/M)} \quad : \quad \left[\frac{\text{m}}{\text{N}}\right]$$

which matches the units on the ratio of signals:

$$\frac{X(s)}{U(s)} : \frac{[\text{meters}]}{[\text{Newton}]}$$

2.4 Basic properties of transfer functions

- Eqn (14) is something pretty remarkable:

$$\frac{\text{Output LT}}{\text{Input LT}} : \frac{V(s)}{U(s)} = G_{uv}(s) = \frac{1/M}{s+b/M} \qquad (14, \text{repeated})$$

- It is the ratio of the output LT over the input LT
  - It depends only on the parameters of the system $\{M,\ b\}$, and is independent of any particular signals $\{V(s), U(s)\}$
- Only linear systems have transfer functions
- Differential equations, such as Eqn (7), are said to be in ***time domain***, the corresponding equation of Laplace transforms, such as Eqn (12), are said to be in ***frequency domain***
- The transfer function of a linear system comes from the governing differential equation, and is a ***rational polynomial*** (a ratio of polynomials). Here is another example:

$$\begin{aligned} 5\dddot{y}(t) + 6\ddot{y}(t) + 7\dot{y}(t) + 8y(t) &= 2\ddot{x}(t) + 3\dot{x}(t) + 4x(t) \qquad (21) \\ \iff G_{xy}(s) = \frac{Y(s)}{X(s)} &= \frac{2s^2+3s+4}{5s^3+6s^2+7s+8} \end{aligned}$$

- When a transfer function has a fraction within the fraction, we always multiply out the fractions within the fraction to put the TF in rational polynomial form. For example, consider $G_{xy}(s)$ given by

$$G(s) = \frac{2s+3}{(4s+5)+\left(\frac{s+1}{s+2}\right)} \tag{22}$$

we multiply by $(s+2)$ over $(s+2)$ to give:

$$\begin{aligned} G(s) &= \frac{2s+3}{(4s+5)+\left(\frac{s+1}{s+2}\right)} = \frac{2s+3}{(4s+5)+\left(\frac{s+1}{s+2}\right)}\frac{(s+2)}{(s+2)} = \frac{(2s+3)(s+2)}{(4s+5)(s+2)+(s+1)} \\ &= \frac{2s^2+7s+6}{4s^2+14s+11} \quad \text{(This is rational polynomial form)} \end{aligned} \tag{23}$$

- Controls is full of algebraic transformations, such as the steps above.
  - You must do the algebra very carefully !
- We can consider the two polynomials individually

$$G_{xy}(s) = \frac{\text{Numerator polynomial}}{\text{Denominator polynomial}} = \frac{B(s)}{A(s)} = \frac{2s^2+3s+4}{5s^3+6s^2+7s+8}$$

where $B(s)$ and $A(s)$ are not considered as Laplace transforms of signals, but simply as polynomials. As polynomials,

  - $B(s)$ and $A(s)$ each have roots. The roots of the numerator polynomial are called the ***zeros***, and are the solutions to

$$B(s) = 2s^2+3s+4 = 0\,, \qquad \text{zeros: } -0.75+j1.20\,, -0.75-j1.20$$

  - The roots of the denominator polynomial are called the ***poles***, and are the solutions to

$$A(s) = 5s^3+6s^2+7s+8 = 0\,, \quad \text{poles: } -1.171\,, -0.014+j1.168\,, -0.014-j1.168$$

  - $B(s)$ and $A(s)$ each have an order (2 and 3 in this case). The order of $B(s)$ is the number of zeros, the order of $A(s)$ is the number of poles.

2.5 Transfer functions of closed-loop systems

- We'll cover transfer functions and block diagrams in more detail when we cover Franklin *et al.* section 3.2 (7th ed.).
- Here the basics for feedback loops are introduced (this will be useful in Lab part 2). Consider the closed-loop system of figure 3

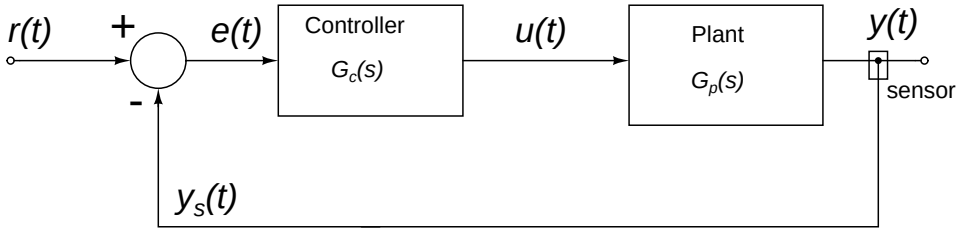


Figure 3: Block diagram with feedback.

- With the feedback,

$$e(t) = r(t) - y(t)$$

so we can write:

$$E(s) = R(s) - Y(s) \tag{24}$$
$$Y(s) = G_p(s)\,U(s) = G_p(s)\,G_c(s)\,E(s) \tag{25}$$

- Combining Eqns (24) and (25), (and omitting (s) on the TF's) we have

$$Y(s) = G_p G_c R(s) - G_p G_c Y(s) \tag{26}$$

- Now, bringing all terms in the output $Y(s)$ to the left-hand side,

$$(1 + G_p G_c)\,Y(s) = G_p G_c R(s) \tag{27}$$

which gives the closed-loop transfer function:

$$\frac{Y(s)}{R(s)} = T_{ry}(s) = \frac{G_c G_p}{1 + G_c G_p} \tag{28}$$

- Eqn (28) is a special case of the general form:

$$T(s) = \frac{\text{Forward path gain}}{1 + \text{Loop gain}} \tag{29}$$

2.5.1 Second example, find $T_{ru}(s)$

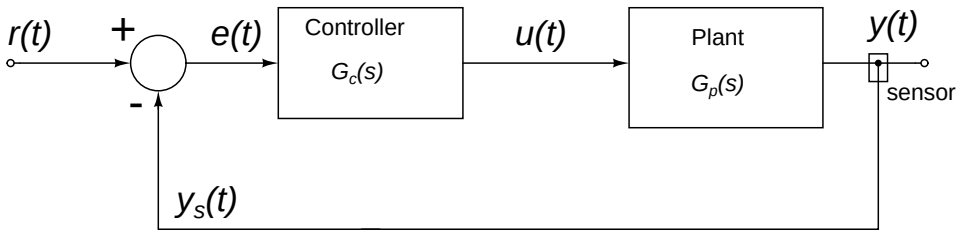


Figure 4: Block diagram with feedback (repeated).

- As a second example, find $T_{ru}(s) = \frac{U(s)}{R(s)}$.
  - Note that in this calculation we are finding the transfer function from $r(t)$ to signal $u(t)$
- Considering the block diagram, we have

$$\begin{aligned} E(s) &= R(s) - Y(s) \\ U(s) &= G_c E(s) \\ Y(s) &= G_p U(s) \end{aligned} \tag{30}$$

  So we can write

$$E(s) = R(s) - G_p U(s) \tag{31}$$

  which gives

$$U(s) = G_c\left(R(s) - G_p U(s)\right) = G_c R(s) - G_c G_p U(s) \tag{32}$$

- Bring all terms with the output to the left-hand side gives,

$$\left(1 + G_c G_p\right) U(s) = G_c R(s) \tag{33}$$

  which gives the transfer function

$$T_{ru}(s) = \frac{G_c}{1 + G_c G_p} \tag{34}$$

- And once again, the closed-loop transfer function is given as

$$T_{ru}(s) = \frac{\text{Forward path gain}}{1 + \text{Loop gain}} \tag{35}$$

2.6 Root locus form and Bode form, two standard forms for the transfer function

- Consider the transfer function

$$G_{uy}(s) = \frac{2s+4}{3s^2+5s+7} \tag{36}$$

There are two ways we can pull a constant out of $G_{uy}(s)$:

2.6.1 Transfer function in root-locus form

- Pull out a leading constant to make the leading coefficients 1.0 (root locus form):

$$G_{uy}(s) = K_{rlg}\,\frac{s+b_0}{s^2+a_1 s+a_0} \tag{37}$$

- This is called the "***root-locus form***," and is used with the inverse Laplace transform and with the root locus. For example (36), we have

$$G_{uy}(s) = \frac{2}{3}\,\frac{1.0s+2}{1.0s^2+5/3s+7/3}$$

and so K_{rlg} =2/3.

2.6.2 Transfer function in Bode form

- Make the DC gain of the rational polynomial component equal to 1.0. This takes two steps:

  1. Pull out any poles or zeros at the origin, into the term $1/s^{N_o}$,

  2. Pull out a leading constant, that makes the b_0 and a_0 coefficients equal to 1.0:

$$G_{uy}(s) = K_{Bode}\,\frac{1}{s^{N_o}}\,\frac{b_1 s + 1.0}{a_2 s^2 + a_1 s + 1.0} \tag{38}$$

  This is called the "***Bode form***". The Bode form makes explicit the DC gain (when $N_o = 0$), and is used to look at the frequency response of systems with the Bode plot (more later). For example, we have

$$G_{uy}(s) = \frac{4}{7}\,\frac{1}{s^0}\,\frac{\frac{1}{2}s + 1}{\frac{3}{7}s^2 + \frac{5}{7}s + 1}$$

  and so $N_o = 0$ and $K_{Bode} = K_{DC} = 4/7$.

2.7 Summary on transfer functions

- The transfer function comes directly from the differential equation, for example
$$3\ddot{y}(t) + 5\dot{y}(t) + 7y(t) = 2\dot{u}(t) + 4u(t)$$
gives the TF
$$G_{uy}(s) = \frac{2s+4}{3s^2+5s+7}$$

- The transfer function is considered a generalized gain, it is the ratio of
$$\frac{\text{Out put Laplace Transform}}{\text{Input Laplace Transform}}$$

- Closing the loop gives a new transfer function, given as
$$T_{ry}(s) = \frac{\text{Forward path gain}}{1 + \text{Loop gain}}$$

- We've seen three standard forms for transfer functions :
  - Rational polynomial form
  - Root locus form
  - Bode form

3 Constituent and continuity equations

- Back to modeling, in general, dynamic systems are described by two types of equations:

1. ***Constituent equations***, these are equations that describe individual elements:

   - Mechanical systems:

$$\text{Spring:} \quad f(t) = k_s x(t) \qquad \text{Mass:} \quad f(t) = m\frac{d^2x(t)}{dt^2}$$
$$\text{Damper:} \quad f(t) = b\frac{dx(t)}{dt}$$

   - Electrical systems:

$$\text{Resistor:} \quad V_R(t) = I_R(t)\,R \qquad \text{Inductor:} \quad V_L(t) = L\frac{dI_L(t)}{dt}$$
$$\text{Capacitor:} \quad I_C(t) = C\frac{V_C(t)}{dt}$$

2. ***Continuity equations***, these are equations that describe the interconnections of the system,

   - Mechanical systems:
     - The sum of the forces on a body (including the inertial force) equals zero (Newton's second law)
     - The sum of relative velocities around any closed loop is equal to zero.
   - Electrical systems (Kirchhoff's laws):
     - Sum of currents leaving a node equals zero
     - Sum of voltages around a loop equals zero.

Many types of dynamic systems can be analyzed in terms of constituent and continuity equations

- Mechanical
- Fluid
- Thermal
- Electric circuits
- Electro-Mechanical systems (motors)

3.1 Procedure to develop dynamic modeling equations

1. Setup model
   (a) Draw a schematic diagram
   (b) Assign variables to needed signals and parameters
      - Mechanical: $x(t)$ or $\theta(t)$ necessary to determine the position of all moving masses
      - Electrical: assign node voltages, and voltage across and current through each element
   (c) Make tables of signals and parameters, including units
   (d) Write the ***constituent equations*** for the elements of the system
2. Write the ***continuity equations*** describing the connections of the system.
   - Mechanical:
     – Draw a free-body diagram of each component, indicate all forces acting on each body
     – Apply Newton's laws, acceleration given by the sum of forces.
   - Electrical
     – Use Kirchhoff's current and voltage laws to analyze the circuit.

   Combine the constituent equations and continuity equations to write the governing differential equations for the system.
3. Put the governing differential equation in standard form (output signal on left, input signals on right)
4. Determine the transfer function(s)
   - Convert differential equations to frequency domain
   - Combine frequency domain equations to eliminate internal variables
   - Often writing differential equations in frequency domain simplifies the algebra
5. Validate that units balance in the equations (governing diff. eqn., or transfer function), to verify correctness

4 Additional mechanical examples

4.1 Example 2.2 (7th ed.), Quarter vehicle suspension

Step 1 Setup model

- Draw a schematic diagram

1/4 Vehicle:

Wheel:

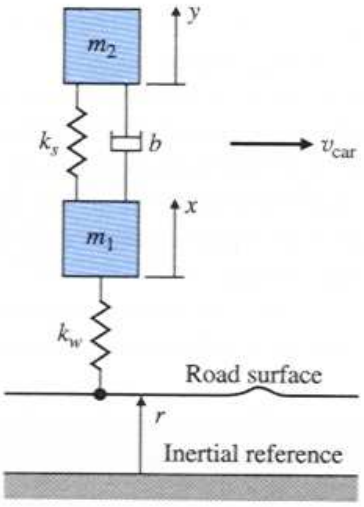


Figure 5: Car Suspension, schematic.

  - Assign variables to signals and parameters (in schematic)

Continuing step 1

– Make lists of signals and parameters

Signal	Units	Description
$x(t)$	[m]	Position of tire center
$y(t)$	[m]	Position of car frame
$r(t)$	[m]	Road height (depends on $v_{car}(t)$)

Parameter	Units	Description
k_w	[N/m]	Tire Stiffness
k_s	[N/m]	Spring Stiffness
b	[N-s/m]	Shock Absorber
M_1	[Kg]	Mass of Tire
M_2	[Kg]	$\frac{1}{4}$ Mass of Car

– Write the constituent equation for each element of the system. Five elements give five equations:

$$\begin{aligned} f_{k_w}(t) &= k_w(r(t)-x(t)) \\ f_{k_s(t)} &= k_s(y(t)-x(t)) \\ f_b(t) &= b(\dot{y}(t)-\dot{x}(t)) \\ m_1\ddot{x}(t) &= f_{net-m_1}(t) \\ m_2\ddot{y}(t) &= f_{net-m_2}(t) \end{aligned}$$

Step 2 Write the continuity equations describing the mechanical system

- Draw the free-body diagram for each mass

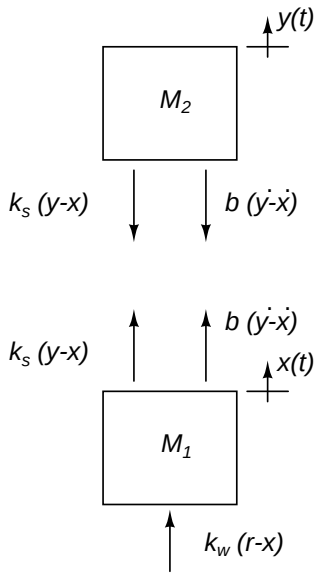


Figure 6: Free-body diagram, shows all forces acting on each body.

- Apply Newton's laws, acceleration given by the sum of forces

  Mass 1:

$$\begin{aligned} m_1 \ddot{x}(t) &= f_b(t) + f_{k_s}(t) + f_{k_w}(t) \\ &= b(\dot{y}(t) - \dot{x}(t)) + k_s(y(t) - x(t) + k_w(r(t) - x(t)) \end{aligned}$$

  Mass 2:

$$\begin{aligned} m_2 \ddot{y}(t) &= -f_{k_s}(t) - f_b(t) \\ &= -b(\dot{y}(t) - \dot{x}(t)) - k_s(y(t) - x(t)) \end{aligned}$$

Step 3 Putting the two differential equations in standard form:

$$m_1\,\ddot{x}(t)+b\,\dot{x}(t)+(k_s+k_w)\,x(t) = b\,\dot{y}(t)+k_s\,y(t)+k_w\,r(t) \quad (39)$$

$$m_2\,\ddot{y}(t)+b\,\dot{y}(t)+k_s\,y(t) = b\,\dot{x}(t)+k_s\,x(t) \quad (40)$$

Step 4 Determine the transfer function

- Write the governing differential equations in frequency domain:

$$\left(m_1\,s^2+b\,s+(k_s+k_w)\right)X(s) = (b\,s+k_s)\,Y(s)+k_w\,R(s) \quad (41)$$

$$\left(m_2\,s^2+b\,s+k_s\right)Y(s) = (b\,s+k_s)\,X(s) \quad (42)$$

- Perform algebraic steps to eliminate internal variables. For example, to derive $Y(s)$ from $R(s)$, eliminate $X(s)$ from Eqns (41) and (42) by using Eqn (42) to write

$$X(s)=\frac{m_2\,s^2+b\,s+k_s}{b\,s+k_s}\,Y(s) \quad (43)$$

Plugging this into Eqn (41) gives

$$\left(m_1\,s^2+b\,s+(k_s+k_w)\right)\frac{m_2\,s^2+b\,s+k_s}{b\,s+k_s}\,Y(s)=(b\,s+k_s)\,Y(s)+k_w\,R(s) \quad (44)$$

Clearing the fraction (almost always a good step) gives

$$\begin{aligned}&\left(m_1\,s^2+b\,s+(k_s+k_w)\right)\left(m_2\,s^2+b\,s+k_s\right)Y(s)\\&\quad=(b\,s+k_s)\,(b\,s+k_s)\,Y(s)+(b\,s+k_s)\,k_w\,R(s)\end{aligned}$$

And bringing all terms in $Y(s)$ (output) to the left, and all terms in $R(s)$ (input) to the right gives

$$\begin{aligned}&\left(\left(m_1\,s^2+b\,s+(k_s+k_w)\right)\left(m_2\,s^2+b\,s+k_s\right)-(b\,s+k_s)\,(b\,s+k_s)\right)Y(s)\\&\quad=(b\,s+k_s)\,k_w\,R(s) \quad (45)\end{aligned}$$

- Now, multiply out the polynomials. This can be done by forming each product term and keeping track of the power of s:

s^4	s^3	s^2	s^1	s^0
$m_1 m_2$	$m_1 b$	$m_1 k_s$	$b k_s$	$k_s (k_s + k_w)$
	$m_2 b$	$m_2 (k_s + k_w)$	$b (k_s + k_w)$	
		b^2		
		$-b^2$	$-b k_s - b k_s$	$-k_s^2$

Where the negative terms come from the which gives

$$G_{ry(s)} = \frac{Y(s)}{R(s)} = \frac{(b s + k_s) k_w}{m_1 m_2 s^4 + (m_1 + m_2) b s^3 + ((m_1 + m_2) k_s + m_2 k_w) s^2 + b k_w s + k_s k_w} \tag{46}$$

Note: deriving transfer functions can involve considerable algebra.

- Take one step at a time, and take several steps
- This is just the quarter vehicle (imagine the whole thing). Computer algebra is often used today
- State-variable modeling (advanced topic, in EE/ME 574 or EE/ME 701) is an approach without so much algebra
- Who is more likely to keep their job: the engineer who can derive four transfer functions in a day, or the engineer who can derive two transfer functions in a day, but knows they are correct
- Double-check often !

Step 5 Validate that units balance

- We can check units in any engineering equation. In this analysis, candidates are the differential equation, frequency domain equations or final transfer function.
  - For a complicated derivation (such as this one) validating each step is not a bad idea.
- Thinking for a moment about what the units of $G_{ry}(s)$ must be,

$$G_{ry}(s): \frac{\text{Output}}{\text{Input}}: \quad \frac{\text{meters}}{\text{meters}}$$

  Transfer function $G_{ry}(s)$ is dimensionless. So our validation of units must give dimensionless.

- Looking at Eqn (46), the units are

$$G_{ry}(s): \quad \frac{\text{Kg}^2 s^{-4} + \text{Kg}\frac{\text{N s}}{\text{m}} s^{-3} + \text{Kg}\frac{\text{N}}{\text{m}} s^{-2} + \frac{\text{N s}}{\text{m}}\frac{\text{N}}{\text{m}} s^{-1} + \frac{\text{N}}{\text{m}}\frac{\text{N}}{\text{m}}}{\left(\frac{\text{N s}}{\text{m}} s^{-1} + \frac{\text{N}}{\text{m}}\right)\frac{\text{N}}{\text{m}}}$$

  - Utilizing

$$[\text{Kg}] = \left[\frac{\text{N s}^2}{\text{m}}\right]$$

    we get

$$G_{ry}(s): \quad \frac{\frac{\text{N}^2\text{s}^4}{\text{m}^2} s^{-4} + \frac{\text{N s}^2}{\text{m}}\frac{\text{N s}}{\text{m}} s^{-3} + \frac{\text{N s}^2}{\text{m}}\frac{\text{N}}{\text{m}} s^{-2} + \frac{\text{N s}}{\text{m}}\frac{\text{N}}{\text{m}} s^{-1} + \frac{\text{N}}{\text{m}}\frac{\text{N}}{\text{m}}}{\left(\frac{\text{N s}}{\text{m}} s^{-1} + \frac{\text{N}}{\text{m}}\right)\frac{\text{N}}{\text{m}}} \tag{47}$$

$$G_{ry}(s) \; : \qquad \frac{\frac{\mathrm{N}^2\mathrm{s}^4}{\mathrm{m}^2}s^{-4} + \frac{\mathrm{N\,s}^2}{\mathrm{m}}\frac{\mathrm{N\,s}}{\mathrm{m}}s^{-3} + \frac{\mathrm{N\,s}^2}{\mathrm{m}}\frac{\mathrm{N}}{\mathrm{m}}s^{-2} + \frac{\mathrm{N\,s}}{\mathrm{m}}\frac{\mathrm{N}}{\mathrm{m}}s^{-1} + \frac{\mathrm{N}}{\mathrm{m}}\frac{\mathrm{N}}{\mathrm{m}}}{\left(\frac{\mathrm{N\,s}}{\mathrm{m}}s^{-1} + \frac{\mathrm{N}}{\mathrm{m}}\right)\frac{\mathrm{N}}{\mathrm{m}}}$$

- There are many cancellations.
  - Seconds cancel everywhere.
- In the numerator, each term in the sum has units $\frac{\mathrm{N}^2}{\mathrm{m}^2}$. The full expression can be reduced to

$$G_{ry}(s) \; : \qquad \frac{\frac{\mathrm{N}^2}{\mathrm{m}^2} + \frac{\mathrm{N}^2}{\mathrm{m}^2} + \frac{\mathrm{N}^2}{\mathrm{m}^2} + \frac{\mathrm{N}^2}{\mathrm{m}^2} + \frac{\mathrm{N}^2}{\mathrm{m}^2}}{\frac{\mathrm{N}^2}{\mathrm{m}^2} + \frac{\mathrm{N}^2}{\mathrm{m}^2}}$$

- Each term in the sums have identical units (as required) and the full expression is

$$G_{ry}(s) \; : \qquad \left[\frac{\mathrm{N}^2}{\mathrm{m}^2}\right] \; / \; \left[\frac{\mathrm{N}^2}{\mathrm{m}^2}\right] = [\text{dimensionless}]$$

- Units balance !

4.2 Franklin *et al.* example 2.3 (7th ed.), Rotational motion: satellite control

- Derive the model for satellite attitude control using a gas jet

Step 1 setup model

- Draw a schematic diagram

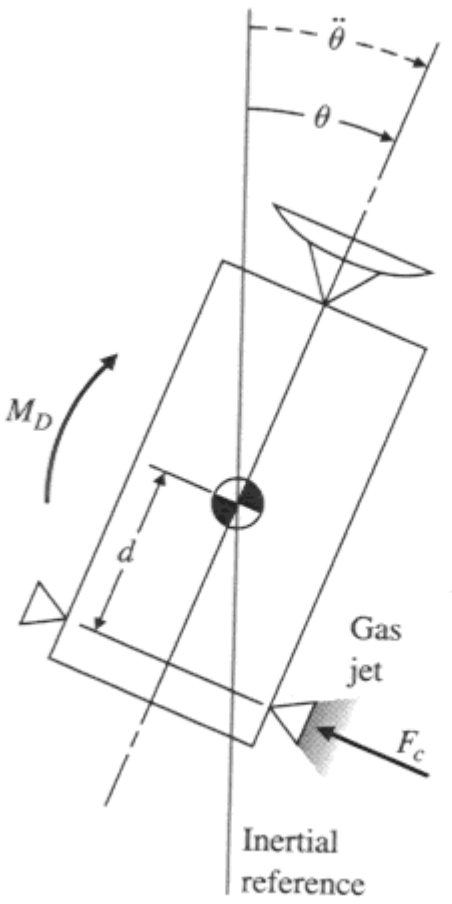


Figure 7: Satellite attitude control, schematic diagram.

- Assign variables to signals and parameters (in figure 7)

- Make tables of signals and parameters

Signal	Units	Description
$\Theta(t)$	[radians]	Orientation of satellite
$F_c(t)$	[N]	Thrust (force)
$M_D(t)$	[N-m]	Disturbance (torque)

Table 5: Signals.

Parameter	Units	Description
I	[Kg-m^2]	Inertia
d	[m]	Thruster radius

Table 6: Parameters.

- Write the constituent equations for the elements of the system. The system only has the rotating inertia, which is governed by:

$$I\,\ddot{\theta}\,(t) = M_{net}\,(t) \tag{48}$$

Step 2 Write the continuity equations describing the rotational mechanical system

- Draw the free-body diagram for each inertia

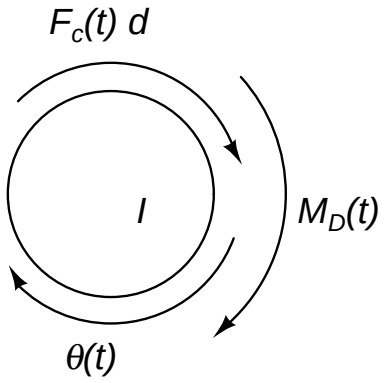


Figure 8: Free-body diagram for satellite.

- Apply Newton's laws, acceleration given by the sum of moments

$$\begin{aligned} M_{net}(t) &= F_c(t)\,d + M_D(t) \\ I\,\ddot{\theta}(t) &= F_c(t)\,d + M_D(t) \end{aligned} \tag{49}$$

Step 3 Putting the differential equation in standard form:

- Eqn (49) is already in standard form

Step 4 Determine the transfer function (student exercise)

Step 5 Validate that units balance

- Checking units of the differential equations, and recalling the units of basic mechanical terms, we find:

$$\left[\frac{\text{N-m-s}^2}{\text{radian}}\right]\left[\frac{\text{radian}}{\text{second}^2}\right] = [\text{N}]\,[\text{m}] + [\text{N-m}]$$

  Which gives

$$[\text{N-m}] = [\text{N}]\,[\text{m}] + [\text{N-m}]$$

- Units balance (and this equation is not dimensionless).

4.3 Example 2.4 flexibility: flexible read/write head for a disk drive

Step 1 Setup the model

- Draw a schematic diagram

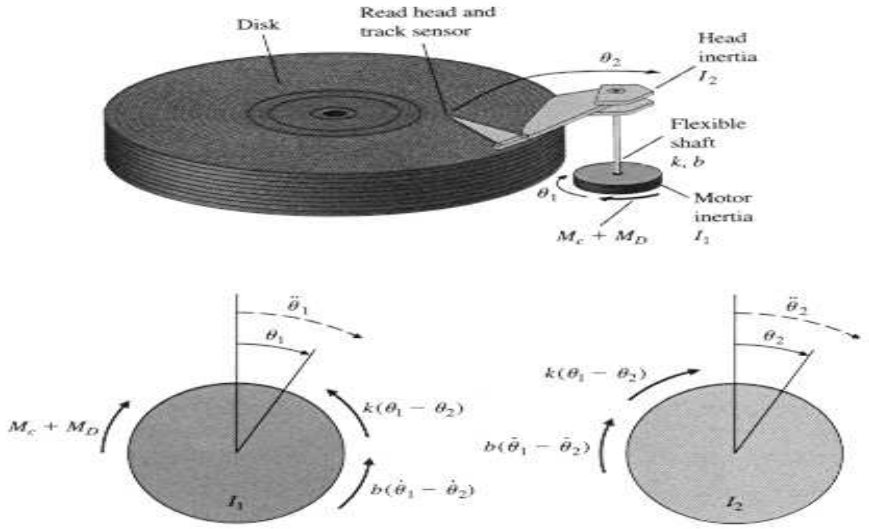


Figure 9: Disk Head Motion, schematic and free body diagram.

For this mechanism:

- There is a spring force: $f_k(t) = k\left(\theta_1(t) - \theta_2 t\right)$
- There is a damping (friction) force: $f_b(t) = b\left(\dot{\theta}_1(t) - \dot{\theta}_2 t\right)$

- Assign variables to signals and parameters (in figure 9)

- Make tables of signals and parameters

Signal	Units	Description
$\theta_1(t)$	[rad]	Position of motor
$\theta_2(t)$	[rad]	Position of disk head
$M_c(t)$	[N-m]	Motor Torque
$M_D(t)$	[N-m]	Disturbance Torque

Table 7: Signals for disk head control.

Parameter	Units	Description
I_1	[Kg-m^2]	Motor Inertia
I_2	[Kg-m^2]	Head Inertia
k	[N-m/rad]	Shaft Stiffness
b	[N-m-s/rad]	Shaft Damping

Table 8: Parameters for disk head control.

- Write the constituent equations for the elements of the system. We have 4 elements:

$$\begin{aligned} \text{Motor inertia:} \quad & I_1 \ddot{\theta}_1(t) = M_{net-\theta_1}(t) \\ \text{Head inertia:} \quad & I_2 \ddot{\theta}_2(t) = M_{net-\theta_2}(t) \\ \text{Shaft stiffness:} \quad & M_k(t) = k\left(\theta_1(t) - \theta_2(t)\right) \\ \text{Shaft flex damping:} \quad & M_b(t) = b\left(\dot{\theta}_1(t) - \dot{\theta}_2(t)\right) \end{aligned} \tag{50}$$

Step 2 Write the continuity equations describing the system

- Draw the free-body diagram (see figure 9)
- Net moment on each inertia. From the free-body diagram:

$$\begin{aligned} M_{net-\theta_1}(t) &= M_c + M_D - M_k(t) - M_b(t) \\ M_{new-\theta_2}(t) &= M_k(t) + M_b(t) \end{aligned} \tag{51}$$

  Notice the signs of the moments in the net moment.

  - The net moment is always in the same sense as $\theta_i(t)$ is measured.
  - Each moment then makes a positive or negative contribution according to its direction relative to $M_{net}(t)$.

- Combine the constituent and continuity equations to write the governing differential equations for the system. Using Eqns (50) and (51), the differential equations are

$$\begin{aligned} I_1\ddot{\theta}_1 = M_{net-\theta_1}(t) &= M_c + M_D - k(\theta_1 - \theta_2) - b\left(\dot{\theta}_1 - \dot{\theta}_2\right) \\ I_2\ddot{\theta}_2 = M_{net-\theta_2}(t) &= k(\theta_1 - \theta_2) + b\left(\dot{\theta}_1 - \dot{\theta}_2\right) \end{aligned}$$

Step 3 Put governing differential equations in standard form

$$\begin{aligned} I_1\ddot{\theta}_1 + b\dot{\theta}_1 + k\theta_1 &= M_c + M_D + b\dot{\theta}_2 + k\theta_2 \\ I_2\ddot{\theta}_2 + b\dot{\theta}_2 + k\theta_2 &= b\dot{\theta}_1 + k\theta_1 \end{aligned} \tag{52}$$

Step 4 Determine the transfer function

- One procedure:

  1. Consider the goal, deriving:

$$\frac{\text{Output}}{\text{Input}}: \qquad \frac{\theta_2(s)}{M_c(s)}$$

  2. Take the Laplace transform of both sides of each governing differential equation

$$\begin{aligned} I_1 s^2 \theta_1(s) + b s \theta_1(s) + k \theta_1(s) &= M_C(s) + M_D(s) + b s \theta_2(s) + k \theta_2(s) \\ I_2 s^2 \theta_2(s) + b s \theta_2(s) + k \theta_2(s) &= b s \theta_1(s) + k \theta_1(s) \end{aligned} \tag{53}$$

  3. Factor each side, pulling out the Laplace transforms

  4. Solve for the desired output from the desired input

     (a) Algebraically eliminate internal variables. Here we can use

$$\theta_1(s) = \frac{I_2 s^2 + b s + k}{b s + k} \theta_2(s) \tag{54}$$

     (b) Clear fractions within the fraction

     (c) Multiply out polynomials

     (d) Simplify

Step 5 Validate units

- Checking units does not guarantee correctness, but checking units will catch many types of errors.

5 Models of electric circuits, (Franklin *et al.* section 2.2)

- Constituent equations

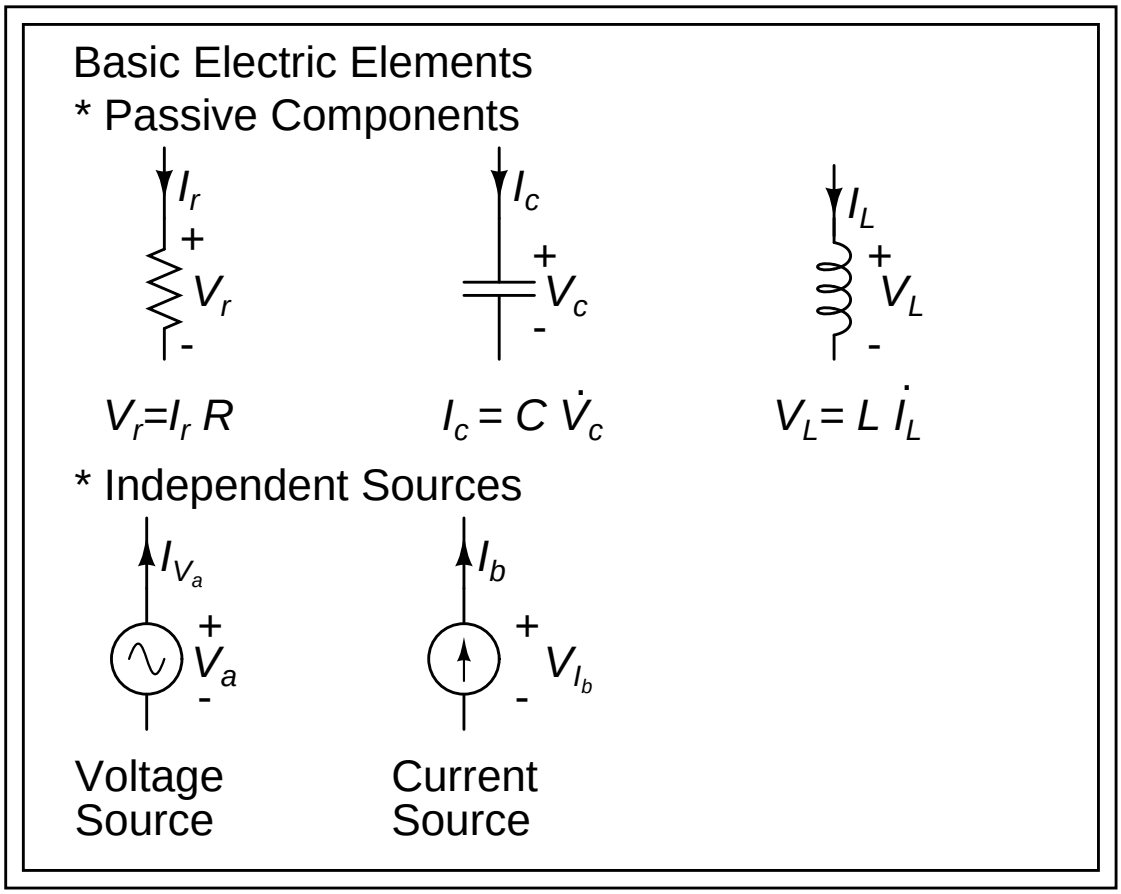


Figure 10: Basic circuit elements provide the constituent equations.

- Continuity Equations, (Kirchhoff's laws)

$$\text{KCL}: \quad \sum_i I_i = 0 \quad \text{Out of a node} \tag{55}$$

(Current positive in equation if the current arrow is pointing out of the node.)

$$\text{KVL}: \quad \sum_i V_i = 0 \quad \text{Around a loop} \tag{56}$$

(Voltage positive in equation if loop path touches the component at '+' side.)

5.1 First circuit example

Step 1 Setup model

- Draw schematic diagram

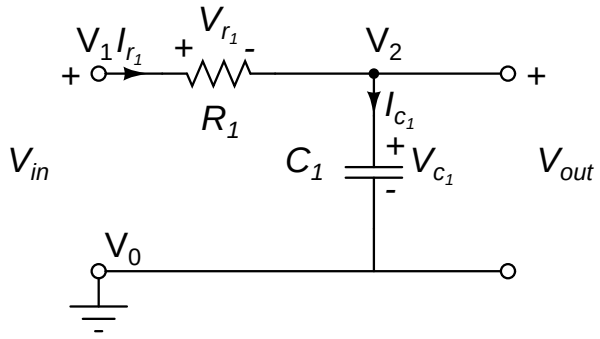


Figure 11: Schematic diagram for a simple circuit example.

- Assign variables to signals and parameters, see schematic.
  - Note application of passive sign convention for electrical circuits: passive element current measurement directed in at '+' terminal of element voltage measurement.

- Make tables of signals and parameters

Signal	Units	Description	Signal	Units	Description
V_0	volts	Ground node	$I_{r1}(t)$	amps	R_1 current
$V_1(t)$	volts	Node 1 voltage	$V_{r1}(t)$	volts	R_1 voltage
$V_2(t)$	volts	Node 2 voltage	$I_{C1}(t)$	amps	C_1 current
$V_{in}(t)$	volts	Input voltage	$V_{C1}(t)$	volts	C_1 voltage
$V_{out}(t)$	volts	Output voltage			

Table 9: Signals for electrical circuit of figure 11.

Parameter	Units	Description
R_1	Ohms: $\left[\frac{v}{A}\right]$	Resistance
C_1	$\left[\frac{A}{v/s}\right] = \left[\frac{A-s}{v}\right]$	Capacitance

Table 10: Parameters of the first circuit.

- We get a long list of signals, for even a simple electrical circuit
  - A voltage and a current for every element
  - Node voltages or loop currents
  - Input and output voltages or currents

  This simple two-element circuit gives us 9 signals to consider.

- Passive sign convention gives a systematic approach for getting the signs right, eliminating the need for free-body diagrams.

- Write the constituent equations for the elements of the system. Here we have two elements

$$\begin{aligned} V_{r1}(t) &= R_1 I_{r1}(t) \\ I_{C1}*(t) &= C_1 \frac{dV_{C1}(t)}{dt} \end{aligned}$$

  Additionally we can add the ground connection, and relations that come from the definitions of signals in the schematic diagram

$$\begin{aligned} V_0 &= 0\,[volts] \\ V_1(t) &= V_{in}(t) \\ V_2(t) &= V_{C1}(t) = V_{out}(t) \end{aligned}$$

Step 2 Write the continuity equations describing the connections of the system

Kirchhoff's current law

$$\text{At node 2:} \qquad -I_{r1}(t) + I_{C1}(t) = 0$$

Kirchhoff's voltage law, around the first loop

$$-V_{in}(t) + V_{r1}(t) + V_{c1}(t) = 0$$

- Combine the constituent and continuity equations to derive the governing differential equation.

$$C_1 \dot{V}_{C1}(t) = I_{C1}(t) = I_{r1}(t) = \frac{1}{R_1} V_{r1}(t) \tag{57}$$

which gives

$$C_1 \dot{V}_{C1}(t) = \frac{1}{R_1} \left(V_{in}(t) - V_{C1}(t) \right) \tag{58}$$

Step 3 Put the governing differential equation in standard form

$$\dot{V}_{C1}(t) + \frac{1}{R_1 C_1} V_{C1}(t) = \frac{1}{R_1 C_1} V_{in}(t) \tag{59}$$

Step 4 Determine the transfer function (student exercise)

Step 5 Validating model by verifying units

$$\left[\frac{volts}{sec} \right] + \left[\frac{1}{sec} \right] [volts] = \left[\frac{1}{sec} \right] [volts]$$

5.2 Example 2.6, a more complex circuit

Step 1 Setup model

- Draw schematic diagram

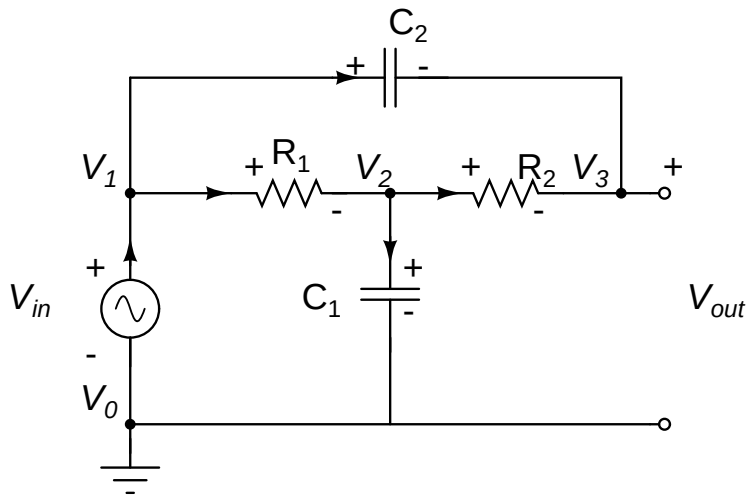


Figure 12: Schematic for example 2.6. Components and the direction of voltage and current measurement are indicated.

- Assign variables to signals and parameters,
  - Note that the measurement direction of **every signal** must be indicated in the schematic, for the meaning of that signal to be **well defined** in equations where it is used.
  - Every device has a voltage across it (e.g., $V_{r1}(t), V_{c2}(t)$) and a current through it (e.g., $I_{r1}(t), I_{c2}(t)$).

- Make tables of signals and parameters

Signal	Units	Description	Signal	Units	Description
V_0	volts	Ground node	$I_{r1}(t)$	amps	R_1 current
$V_1(t)$	volts	Node 1 voltage	$V_{r1}(t)$	volts	R_1 voltage
$V_2(t)$	volts	Node 2 voltage	$I_{r2}(t)$	amps	R_2 current
$V_3(t)$	volts	Node 3voltage	$V_{r2}(t)$	volts	R_2 voltage
			$I_{C1}(t)$	amps	C_1 current
			$V_{c1}(t)$	volts	C_1 voltage
$V_{in}(t)$	volts	Input voltage	$I_{C2}(t)$	amps	C_2 current
$V_{out}(t)$	volts	Output voltage	$V_{c2}(t)$	volts	C_2 voltage

Table 11: Signals for electrical circuit of figure 12.

Parameter	Units	Description
R_{1-2}	Ohms: $\left[\frac{v}{A}\right]$	Resistance
C_{1-2}	$\left[\frac{A}{v/s}\right] = \left[\frac{A-s}{v}\right]$	Capacitance

Table 12: Parameters of the circuit of figure 12.

- Write the constituent equations for the elements of the system

$$\begin{aligned} I_{r_i}(t) &= R_i V_{r_i}(t) \\ I_{C_i}(t) &= C_i \frac{dV_{C_i}(t)}{dt} \end{aligned}$$

Additionally we can add the ground connection, and relations that come from the definitions of signals in the schematic diagram

$$\begin{aligned} V_0 &= 0\ [volts] \\ V_1(t) = V_{in}(t) \qquad & V_{out}(t) = V_3(t) \\ V_{r1}(t) = V_1(t) - V_2(t) \qquad & V_{C_1}(t) = V_2(t) \\ V_{r2}(t) = V_2(t) - V_3(t) \qquad & V_{C_2}(t) = V_1(t) - V_3(t) \end{aligned}$$

Step 2 Write the continuity equations describing the connections of the system

Kirchhoff's current law

$$\begin{aligned} \text{At node 2:} \quad & -I_{r1}(t) + I_{r2}(t) + I_{C1}(t) = 0 \\ \text{At node 3:} \quad & -I_{r2}(t) - I_{C2}(t) = 0 \end{aligned}$$

Kirchhoff's voltage law

$$\begin{aligned} -V_{in}(t) + V_{r1}(t) + V_{C1}(t) &= 0 \\ V_{C2}(t) - V_{r1}(t) - V_{r2}(t) &= 0 \\ -V_{in}(t) + V_{C2}(t) + V_{out}(t) &= 0 \end{aligned}$$

- Combine the constituent and continuity equations to derive the governing differential equations.
  - Identify the differential equations in the constituent equations
  - Do algebra to write the right hand side using
    1. Input signals
    2. Signals governed by differential equations
  - In this example:
    * The differential equations are the two capacitor equations

$$\begin{aligned} \frac{dV_{C_1}(t)}{dt} &= \frac{1}{C_1} I_{C_1}(t) \\ \frac{dV_{C_2}(t)}{dt} &= \frac{1}{C_2} I_{C_2}(t) \end{aligned}$$

    * Write the right-hand-side of each using $V_{in}(t)$, $V_{C_1}(t)$ and $V_{C_2}(t)$

- Using

$$\begin{aligned} V_{out} &= V_{in} - V_{C2} \\ V_{r2} &= V_{C1} - V_{out} = V_{C1} - V_{in} + V_{C2} \end{aligned}$$

  we can write

$$\begin{aligned} C_1 \dot{V}_{C1}(t) &= I_{C1}(t) = I_{r1}(t) - I_{r2}(t) \\ &= \frac{1}{R_1}(V_{in}(t) - V_{C1}(t)) - \frac{1}{R_2}(V_{C1}(t) - V_{in}(t) + V_{C2}(t)) \end{aligned}$$

$$C_2 \dot{V}_{C2}(t) = I_{C2}(t) = -I_{r2}(t) = -\frac{1}{R_2}(V_{C1} - V_{in} + V_{C2})$$

Step 3 Put the governing differential equations in standard form

$$\dot{V}_{C1}(t) + \left(\frac{1}{R_1 C_1} + \frac{1}{R_2 C_1}\right) V_{C1}(t) = \left(\frac{1}{R_1 C_1} + \frac{1}{R_2 C_1}\right) V_{in}(t) - \frac{1}{R_2 C_1} V_{C2}(t) \quad (60)$$

$$\dot{V}_{C2}(t) + \frac{1}{R_2 C_2} V_{C2}(t) = \frac{1}{R_2 C_2} V_{in}(t) - \frac{1}{R_2 C_2} V_{C1}(t) \quad (61)$$

Step 4 Derive the transfer function

- Here we'll take a look at the algebra to derive the transfer function (TF). The TF is the ratio of the output over the input Laplace transform (LT)

$$G_p(s) = \frac{V_{out}(s)}{V_{in}(s)}$$

Using the KVL equation

$$V_{out}(t) = V_{in}(t) - V_{C2}(t) \quad (62)$$

- As we saw in section 2, differential equations (60) and (61) can be written in the frequency domain. To reduce the algebra, create ***lumped terms***, that is, symbols for groups of terms that appear repeatedly

$$A = \left(\frac{1}{R_1 C_1} + \frac{1}{R_2 C_1}\right) \qquad B = \frac{1}{R_2 C_1}$$
$$C = \frac{1}{R_2 C_2}$$

Using A ... C , the differential equations are written in the frequency domain as:

$$\begin{aligned} sV_{C1}(s) + AV_{C1}(s) &= AV_{in}(s) - BV_{C2}(s) \\ sV_{C2}(s) + CV_{C2}(s) &= CV_{in}(s) - CV_{C1}(s) \end{aligned}$$

- Grouping terms on the left, gives

$$(s+A)\,V_{C1}(s) = A\,V_{in}(s) - B\,V_{C2}(s) \quad (63)$$
$$(s+C)\,V_{C2}(s) = C\,V_{in}(s) - C\,V_{C1}(s) \quad (64)$$

  which is a set of two coupled equations.

- From Eqn (62)

$$V_{out}(s) = V_{in}(s) - V_{C2}(s)$$

  so let's derive $V_{C2}(s)/V_{in}(s)$.

- To derive $V_{C2}(s)/V_{in}(s)$, start with Eqn (64)

$$(s+C)\,V_{C2}(s) = C\,V_{in}(s) - C\,V_{C1}(s) \qquad \text{(Eqn 64, repeated)}$$

  we need to replace the term $V_{C1}(s)$ with an expression using $V_{C2}(s)$ and $V_{in}(s)$. Using Eqn (63) :

$$V_{C1} = \frac{A\,V_{in} - B\,V_{C2}}{(s+A)} \quad (65)$$

  (Where the (s) argument is left off. For Frequency Domain equations, all signals are functions of (s)).

- Eqn (64) becomes

$$(s+C)\,V_{C2} = C\,V_{in} - C\frac{A\,V_{in} - B\,V_{C2}}{(s+A)}$$

  This is an expression with only two signals, intermediate output $V_{C2}(s)$ and input $V_{in}(s)$.

- Now re-arrange and resolve the fraction on the right

$$(s+A)(s+C)V_{C2} = (s+A)CV_{in} - CAV_{in} + BCV_{C2}$$

- Simplify with the cancellation, and bring all terms in V_{C2} (the output for this Eqn) to the left

$$(s+A)(s+C)V_{C2} - BCV_{C2} = CsV_{in} \tag{66}$$

- Eqn (66) gives the transfer function from $V_{in}(s)$ to $V_{C2}(s)$

$$\frac{V_{C2}(s)}{V_{in}(s)} = \frac{Cs+0}{s^2+(A+C)s+AC-BC} = \frac{N_{C2}(s)}{D_{C2}(s)} \tag{67}$$

  where $N_{C2}(s)$ and $D_{C2}(s)$ are introduced, to reduce the algebra in the next step.

- Now write an expression for $V_{out}(s)$ the depends only on $V_{out}(s)$ and $V_{in}(s)$

$$V_{out} = V_{in} - V_{C2} = V_{in} - \frac{N_{C2}}{D_{C2}}V_{in} \tag{68}$$

- Eqn (68) has only $V_{out}(s)$ and $V_{in}(s)$, so we are close. The right hand side can be written

$$V_{out} = \left(1 - \frac{N_{C2}}{D_{C2}}\right)V_{in} = \left(\frac{D_{C2}}{D_{C2}} - \frac{N_{C2}}{D_{C2}}\right)V_{in} = \left(\frac{D_{C2}-N_{C2}}{D_{C2}}\right)V_{in} \tag{69}$$

- Expansion of Eqn (69) leads to

$$G_p(s) = \frac{V_{out}(s)}{V_{in}(s)} = \left(\frac{D_{C2} - N_{C2}}{D_{C2}}\right) = \frac{\left(s^2 + (A+C)s + AC - BC\right) - (Cs + 0)}{s^2 + (A+C)s + AC - BC}$$

  Another cancellation occurs

$$G_p(s) = \frac{V_{out}(s)}{V_{in}(s)} = \frac{s^2 + As + AC - BC}{s^2 + (A+C)s + AC - BC}$$

- Plug in the expression for the grouped terms

$$\frac{V_{out}(s)}{V_{in}(s)} = \frac{s^2 + \left(\frac{1}{R_1C_1} + \frac{1}{R_2C_1}\right)s + \left(\left(\frac{1}{R_1C_1} + \frac{1}{R_2C_1}\right) - \frac{1}{R_2C_1}\right)\frac{1}{R_2C_2}}{s^2 + \left(\frac{1}{R_1C_1} + \frac{1}{R_2C_1} + \frac{1}{R_2C_2}\right)s + \left(\left(\frac{1}{R_1C_1} + \frac{1}{R_2C_1}\right) - \frac{1}{R_2C_1}\right)\frac{1}{R_2C_2}}$$

- Another cancellation occurs

$$\frac{V_{out}(s)}{V_{in}(s)} = \frac{s^2 + \left(\frac{1}{R_1C_1} + \frac{1}{R_2C_1}\right)s + \frac{1}{R_1C_1}\frac{1}{R_2C_2}}{s^2 + \left(\frac{1}{R_1C_1} + \frac{1}{R_2C_1} + \frac{1}{R_2C_2}\right)s + \frac{1}{R_1C_1}\frac{1}{R_2C_2}}$$

- (Optional) multiply through by $R_1R_2C_1C_2 / R_1R_2C_1C_2$, to clear the fractions within the fraction

$$\frac{V_{out}(s)}{V_{in}(s)} = \frac{R_1R_2C_1C_2s^2 + (R_2C_2 + R_1C_2)s + 1}{R_1R_2C_1C_2s^2 + (R_2C_2 + R_1C_2 + R_1C_1)s + 1} \tag{70}$$

Step 5 Validating model by verifying units

- Verifying units in differential equations (60) and (61)

$$\begin{aligned}\left[\frac{volts}{sec}\right]+\left[\frac{1}{sec}+\frac{1}{sec}\right][volts] &= \left[\frac{1}{sec}+\frac{1}{sec}\right][volts]-\left[\frac{1}{sec}\right][volts]\\ \left[\frac{volts}{sec}\right]+\left[\frac{1}{sec}\right][volts] &= \left[\frac{1}{sec}\right][volts]-\left[\frac{1}{sec}\right][volts]\end{aligned}$$

  - In this case, all terms are [volts/second] (check!).

- Verify units in the transfer function, Eqn (70)

$$\frac{V_{out}(s)}{V_{in}(s)} : \frac{[s^2]\,[s^{-2}]+[s]\,[s^{-1}]+[\cdot]}{[s^2]\,[s^{-2}]+[s]\,[s^{-1}]+[\cdot]} = [\cdot] = \frac{[\text{volts}]}{[\text{volts}]}$$

  - In this case, all terms are $[\cdot]$. The transfer function is dimensionless (check!).

5.3 Example: op-amp, low-pass filter

Step 1 Setup model

- Draw schematic diagram

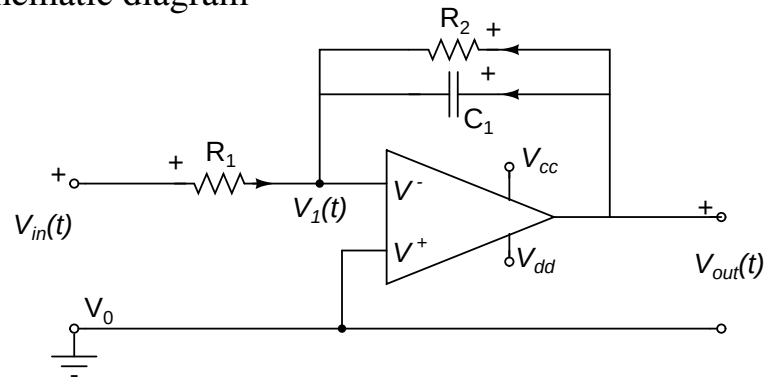


Figure 13: Op amp circuit, low-pass filter. Notes: For the op-amp with negative feedback $\left(V^{+}-V^{-}\right)\approx 0$; This circuit forms a virtual ground at '-' input.

- Assign all signals and parameters, see schematic.
- Make tables of signals and parameters

Signal	Units	Description	Signal	Units	Description
V_0	volts	Ground node	$I_{r1-2}(t)$	amps	R_1, R_2 current
$V_1(t)$	volts	Node 1 voltage	$V_{r1-2}(t)$	volts	R_1, R_2 voltage
$V_{in}(t)$	volts	Input voltage	$I_{C1}(t)$	amps	C_1 current
$V_{out}(t)$	volts	Output voltage	$V_{C1}(t)$	volts	C_1 voltage

Table 13: Signals for electrical circuit of figure 13.

Parameter	Units	Description
R_{1-2}	Ohms: $\left[\frac{v}{A}\right]$	Resistance
C_1	$\left[\frac{A}{v/s}\right]=\left[\frac{A-s}{v}\right]$	Capacitance

Table 14: Parameters of the circuit of figure 13.

- Write the constituent equations for the elements of the system

$$I_{r_i}(t) = R_i V_{r_i}(t) \tag{71}$$

$$I_{C_i}(t) = C_i \frac{dV_{C_i}(t)}{dt} \tag{72}$$

Op Amp, the output voltage is the necessary voltage so that

$$V^+ = V^- \tag{73}$$

- Additionally we can add the ground connection, and relations that come from the definitions of signals in the schematic diagram

$$V_0 = 0\,[volts]$$

$$V_1(t) = V^-(t) \qquad V_{C_1}(t) = V_{out}(t) - V_1(t)$$

$$V_{r1}(t) = V_{in}(t) - V_1(t) \qquad V_{r2}(t) = V_{out}(t) - V_1(t)$$

Because the V^+ terminal is wired to ground, the op amp creates a ***virtual ground*** at the V^- terminal:

$$V_1(t) = V^-(t) = V^+(t) = 0 \tag{74}$$

Step 2 Write the continuity equations describing the connections of the system

Kirchhoff's current law

$$\text{At node 1:} \quad I_{r1}(t) + I_{r2}(t) + I_{C1}(t) = 0 \tag{75}$$

- Combine the constituent and continuity equations to derive the governing differential equation. Using the constituent Eqns for the other elements

$$I_{r1} = \frac{1}{R_1}(V_{in} - V_1) \tag{76}$$

$$I_{r2} = \frac{1}{R_2}(V_{out} - V_1) \tag{77}$$

$$I_{c1} = \frac{1}{C_1}\frac{d}{dt}(V_{out} - V_1) \tag{78}$$

with the constituent Eqn for the Op Amp

$$V_1(t) = V^-(t) = 0 \tag{79}$$

we can write

$$\frac{V_{in}(t)}{R_1} + \frac{V_{out}(t)}{R_2} + C_1\frac{dV_{out}(t)}{dt} = 0 \tag{80}$$

Step 3 Put the governing differential equation in standard form

$$\dot{V}_{out}(t) + \frac{1}{C_1 R_2}V_{out}(t) = -\frac{1}{C_1 R_1}V_{in}(t) \tag{81}$$

Step 4 Derive the transfer function, replacing d/dt in Eqn (81) with s and factoring out $V_{out}(s)$

$$\left(s + \frac{1}{C_1 R_2}\right) V_{out}(s) = -\frac{1}{C_1 R_1} V_{in}(s) \tag{82}$$

and rearranging to form $G_{OA}(s) = \frac{V_{out}(s)}{V_{in}(s)}$

$$G_{OA}(s) = \frac{V_{out}(s)}{V_{in}(s)} = \frac{-\frac{1}{C_1 R_1}}{s + \frac{1}{C_1 R_2}} = \frac{-R_2}{C_1 R_1 R_2 s + R_1} \tag{83}$$

Note: we often end up with "fractions within the fraction." Multiplying through to resolve these fractions is usually a practical step of simplification.

Step 5 Validating model by verifying units in Eqn (82)

$$\left[\frac{\text{volts}}{\text{sec}}\right] + \left[\frac{1}{\text{sec}}\right][\text{volts}] \quad = \quad \left[\frac{1}{\text{sec}}\right][\text{volts}] \tag{84}$$

6 Models of electro-mechanical systems (Franklin *et al.* section 2.3 (7th ed.))

- Current and magnetic fields interact in 3 ways that are important for electro-mechanical machines.

- Carlson and Gisser, *Electrical Engineering Concepts and Applications,* provide a nicely written presentation of the subject.

6.1 Currents produce a magnetic field

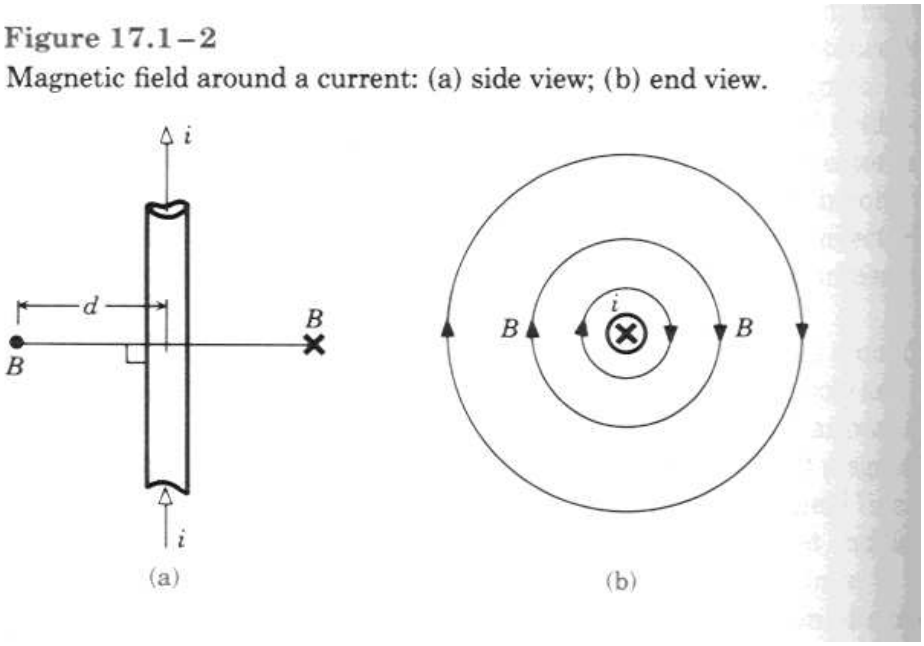


Figure 14: Magnetic field by current (C&G fig 17.1-2).

- Magnetic field strength is proportional to current and inversely proportional to distance:

$$B = \frac{\mu i}{2\pi d} \tag{85}$$

Where:	B	Magnetic Field Strength (Flux Density)	[Tesla]
	μ	Magnetic permeability, $\mu_0 = 4\pi 10^{-7}$ $\mu = \mu_r \mu_0$, $\mu_r \in 1.0...5000$	$\left[\frac{Henries}{meter}\right] = \left[\frac{Newtons}{Amp^2}\right]$
	i	Current	$[Amps]$
	d	Distance	$[meters]$

  - And the direction is given by the right-hand rule.

6.1.1 Units for magnetic fields:

- Magnetic Flux Density is measured in Tesla, A wire of 1 meter with a current of 1 Amp experiences a force of 1 Newton in a field of 1 Tesla. Thus

$$B: \qquad 1\,[\text{Tesla}] = 1\left[\frac{Weber}{m^2}\right] = 1\left[\frac{\text{N}}{\text{A}-\text{m}}\right]$$

  - Considering unites in Eqn (85)

$$[T] = \frac{\left[\frac{N}{A^2}\right][A]}{[m]} = \left[\frac{N}{A-m}\right] \tag{86}$$

- The total magnetic flux is Tesla integrated over an area, and is measured in Webers

$$\Phi: \quad 1\,[\text{Weber}] = 1\left[\text{Tesla m}^2\right] = 1\left[\frac{\text{N}-\text{m}}{\text{Amp}}\right] = 1\left[\frac{\text{Watt-second}}{\text{Amp}}\right] = 1\,[\text{Volt-sec}] \tag{87}$$

- Note on units of power in mechanical and electrical units:

1 Watt	=	1 Newton $\times$ meter/second	(linear motion)
	=	1 Newton-meter $\times$ radian/second	(rotation motion)
	=	1 Volt $\times$ Amp	(electrical)
	=	1 Joule / second	(power is energy per second)

6.1.2 Solenoid example

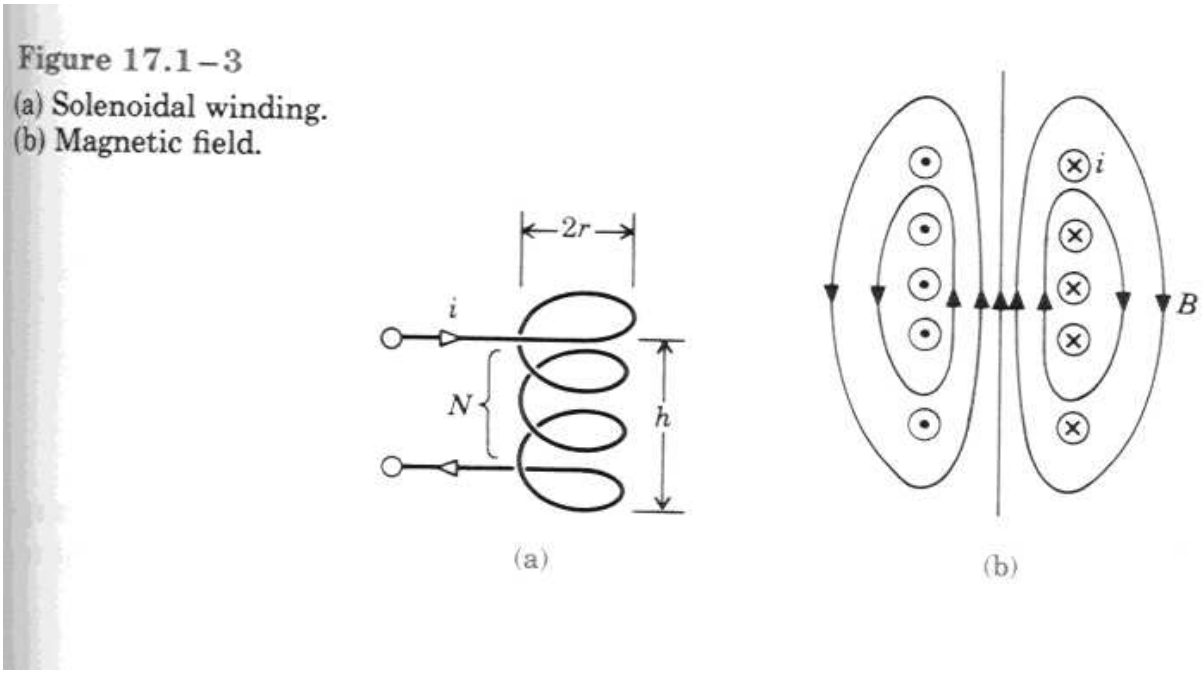


Figure 15: Magnetic field by current in a solenoid (C&G fig 17.1-3).

- Integrating Eqn (85) over the windings gives a field strength within the solenoid of

$$B(t) = \mu \frac{N i(t)}{h} \qquad \left[\frac{N}{Amp^2}\right] \frac{[\cdot]\,[Amps]}{[meter]} = \left[\frac{N}{Amp - meter}\right] \tag{88}$$

where

h: is the solenoid length

N: is the number of turns

i(t) is the current

6.2 Currents moving in a magnetic field experience a force, the Bli law

- The Lorentz force converts electrical energy to mechanical energy.

Figure 17.4–1
Magnetic force on a current-carrying wire.

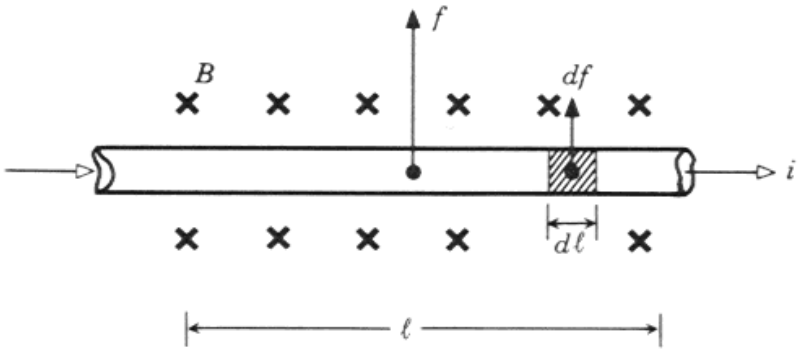


Figure 16: Magnetic force on a conductor (C&G fig 17.4-1)

$$F(t) = B l i(t) \quad [\text{N}] \tag{89}$$

Where:	B	Magnetic Field Strength (Flux Density)	$\left[\frac{\text{Newtons}}{\text{Amp-meter}}\right]$
	l	Length of the conductor	[m]
	i	Current	[A]

Figure 17.1–1 (a) Magnetic field. (b) Force on a moving charge.

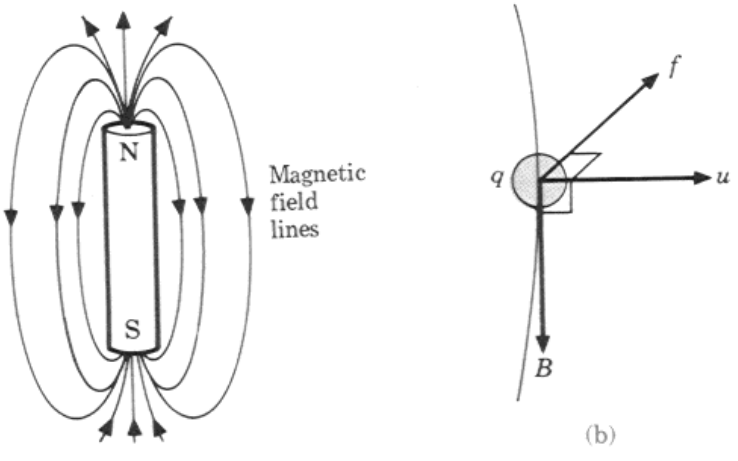


Figure 17: Force on a moving charge (C&G fig 17.1-1).

6.3 Third interaction between current and a magnetic field: the Blu law

$$E_c(t) = B\, l\, u(t) \tag{90}$$

Where:			
	B	Magnetic Field Strength (Flux Density)	[Tesla]=$\left[\frac{\text{Webers}}{\text{m}^2}\right]$
	l	Length of moving wire	[m]
	$u(t)$	Velocity of moving wire (perpendicular to B)	[m/s]
	$E_c(t)$	Electromotive force of coil (voltage)	[volts]

Also: $$e(t) = N_{loops}\frac{d}{dt}\Phi(t) = Area \times N_{loops} \times \frac{d}{dt}B(t) \tag{91}$$

Figure 17.1–5
Voltage induced by time-varying flux.

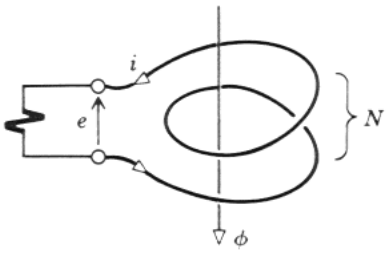


Figure 18: Voltage induced by a time-varying flux, Eqn (91).

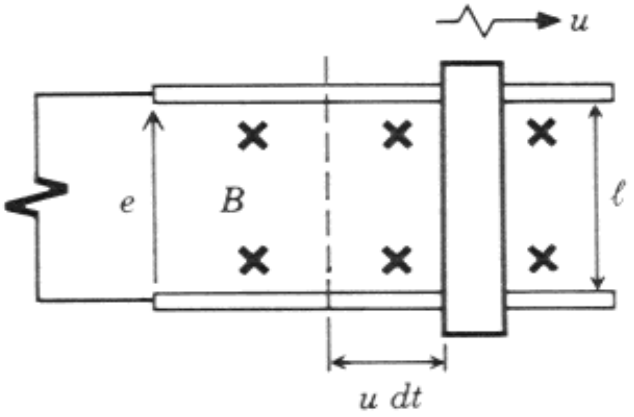


Figure 19: Voltage induced by a moving conductor, Eqn (90).

6.4 Franklin *et al.* Examples 2.11 and 2.12 (7^{th}ed.), Modeling a voice coil motor

- A dynamic model for a motor has two sub-models
  - A model for the electrical side
  - A model for the mechanical side

These result in two model equations, coupled by the force generation,

$$F_c(t) = k_t I_a(t) \ , \qquad k_t : \left[\frac{N}{Amp}\right] \tag{92}$$

and by the back electromotive force (back EMF)

$$E_c(t) = k_e \dot{x}(t) \ , \qquad k_e : \left[\frac{Volts}{m/s}\right] \tag{93}$$

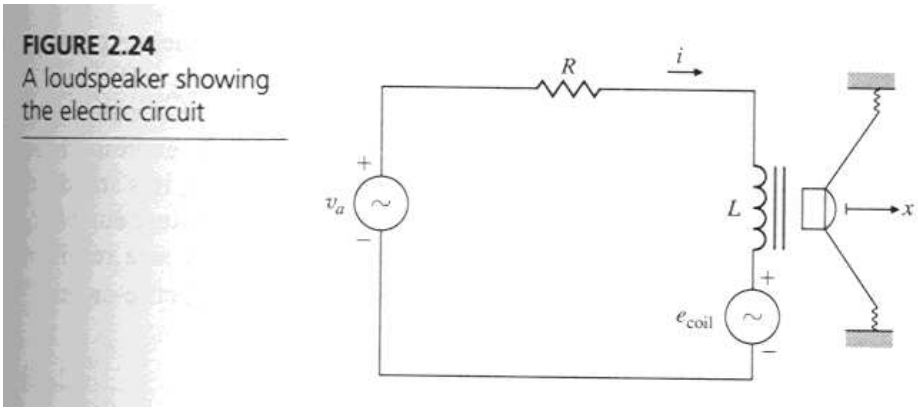


Figure 20: Voice coil motor with equivalent circuit.

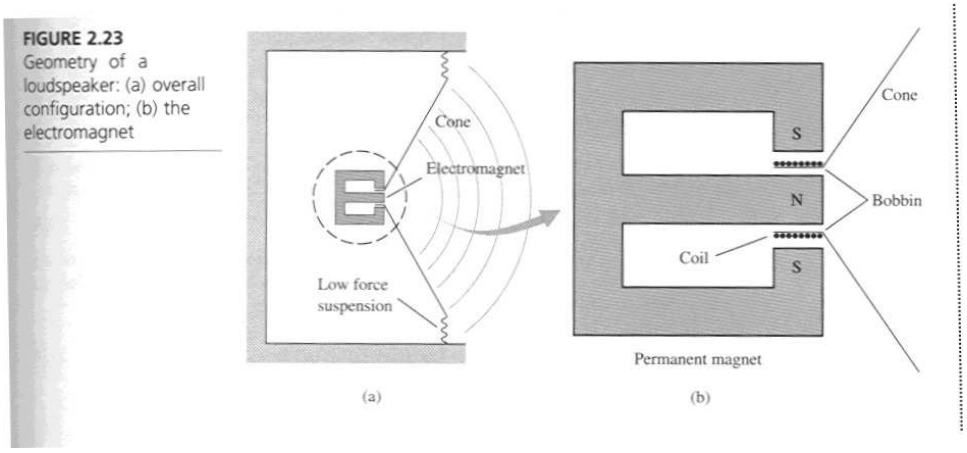
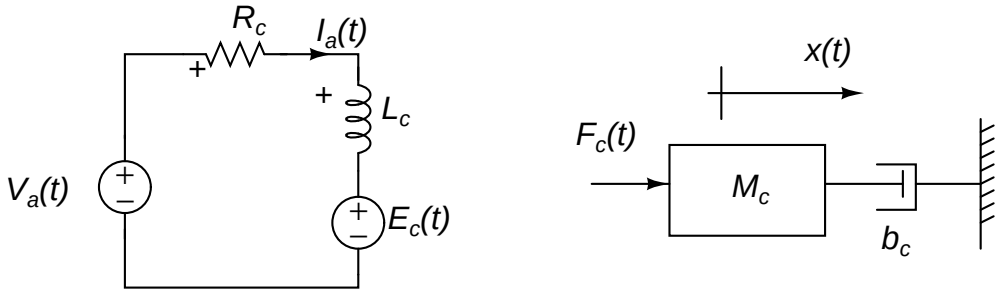


Figure 21: Geometry of a loudspeaker (fig 2.23).

Step 1 Setup model

- Draw a schematic diagram (two parts, for electrical and mechanical sides of motor)



Model, Electrical Elements Model, Mechanical Elements

Figure 22: Schematic diagram of electrical and mechanical elements of a motor.

- Assign variables to signals and parameters, see schematic diagram
- Make tables of signals and parameters

Electrical Side			Mechanical Side		
Signal	Units	Description	Signal	Units	Description
$V_a(t)$	volts	Applied voltage	$x(t)$	meters	Coil position
$I_a(t)$	amps	Armature current	$F_c(t)$	Newtons	Motor force
$E_c(t)$	volts	Back EMF			

Table 15: Signals for motor of figure 22.

Electrical Side			Mechanical Side		
Parameter	Units	Description	Parameter	Units	Description
R_c	Ω	Coil winding resistance	M_c	Kg	Moving mass
L_c	Henries	Coil winding inductance	b_c	$\frac{N}{meter/sec}$	Damping coefficient
k_e	$\frac{volts}{meter/second}$	Back EMF constant	k_t	$\frac{N}{amp}$	Motor force constant

Table 16: Parameters of the motor of figure 22.

- Write constituent equations for the elements of the system

  - Electrical side:

$$\begin{aligned} V_{R_c}(t) &= R_c I_a(t) \\ V_{L_c}(t) &= L_c \frac{d I_a(t)}{dt} \\ E_c(t) &= k_e \dot{x}(t) \end{aligned}$$

  - Mechanical side:

$$\begin{aligned} M_c \ddot{x}(t) &= F_{net.M_c}(t) \\ F_{b_c}(t) &= b_c \dot{x}(t) \\ F_c(t) &= k_t I_a(t) \end{aligned}$$

- Additionally, in this example derive the motor constant from modeling information. The information given is:

  - Bobbin: 20 turns, 2cm diameter,

  - B field in the gap of the permanent magnet: $B = 0.5$ [T]

  - Modeling for linear motor, length of wire in B field:

$$l = (\text{turns})\,(\text{circumference}) = 20\,2\pi \text{ [cm]} = 1.26 \text{ [m]}$$

  - Magnetic Force:

$$F_c(t) = B\,l\,i(t) = 0.5\ 1.26\ I_a(t) = k_t I_a(t) = 0.63\ I_a(t) \quad \text{[N]} \qquad (94)$$

  where $k_t = 0.63$ [Newtons / Amp]

Step 2 Write the continuity equations describing the connections of the system

- Electrical side, KVL:

$$V_{R_c}(t) + V_{L_c}(t) + E_c(t) = V_a(t) \tag{95}$$

- Mechanical side, from free-body diagram

$$M_c \ddot{x}(t) + b_c \dot{x} = F_c(t) \tag{96}$$

- Combine the constituent and continuity equations to derive the governing differential equations.

$$\begin{aligned} L_c \dot{I}_a(t) + R_c I_a(t) &= V_a(t) - k_e \dot{x}(t) && (97) \\ M_c \ddot{x}(t) + b_c \dot{x} &= k_t I_a(t) && (98) \end{aligned}$$

Very high performance systems are realized with voice-coil motors (such as disk drive heads) modeled by Eqn (97) and (98).

Step 3 Put the governing differential equation in standard form

- Eqns (97) and (98) are already in standard form.

Step 4 Derive the transfer function

$$G_{vx}(s) = \frac{X(s)}{V_a(s)}$$

(student exercise)

Step 5 Validating model by verifying units

- Electrical side:

$$\left[\frac{volt-second}{amp}\right]\left[\frac{amp}{sec}\right] + \left[\frac{volt}{amp}\right][amp] \;=\; [volts] - \left[\frac{volt-sec}{meter}\right]\left[\frac{meter}{sec}\right]$$

giving:

$$[volts] + [volts] = [volts] - [volts]$$

- Mechanical side:

$$\left[\frac{N-s^2}{m}\right]\left[\frac{m}{s^2}\right] + \left[\frac{N-s}{m}\right]\left[\frac{m}{s}\right] = \left[\frac{N}{amp}\right][amps]$$

giving

$$[N] + [N] = [N]$$

6.4.1 Motor constants

- A remarkable fact about motors is that,
  - By conservation of energy and when consistent units are used, the back EMF constant equals the motor torque constant :

$$k_e = k_t \tag{99}$$

In our example,

$$k_e = 0.63 \left[\frac{volts}{meter/sec}\right]$$

and

$$k_t = 0.63 \left[\frac{N}{amp}\right]$$

If we consider the ratio of their units

$$\frac{[k_e]}{[k_t]} : \quad \frac{\left[\frac{volts}{meter/sec}\right]}{\left[\frac{N}{amp}\right]} \times \frac{\left[amp - \frac{meter}{sec}\right]}{\left[amp - \frac{meter}{sec}\right]} = \frac{[volt - amp]}{\left[N - \frac{meter}{sec}\right]} = \frac{[Watts]}{[Watts]}$$

- Though they look very different, both k_e and k_t have units of

$$k_e\,,\, k_t \; : \quad \left[\frac{Watt - seconds}{amp - meter}\right] = \left[\frac{\text{Joule}}{\text{Amp-meter}}\right]$$

7 DC motors

- DC motors operate like a linear voice coil motor, except that the motion is rotational.

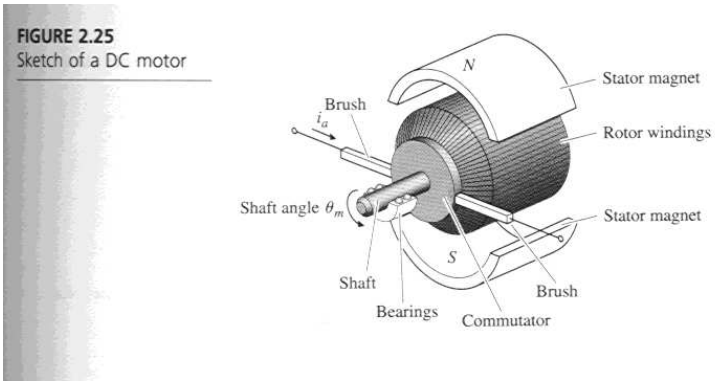


Figure 23: Exploded view of a DC motor (fig 2.25).

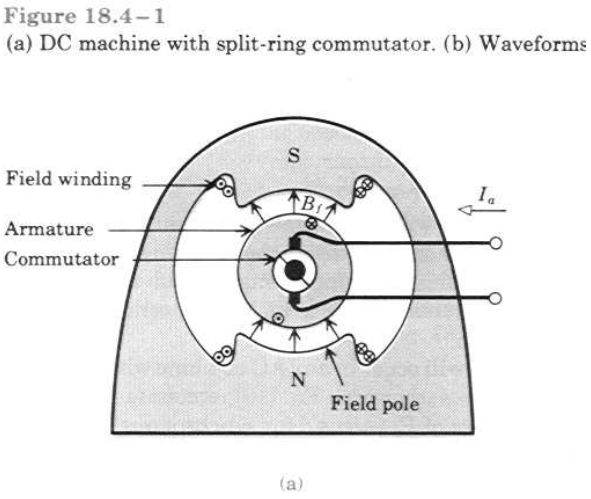


Figure 24: DC machine with commutator (fig 18.4-1).

- DC motor modeling equations (constituent equations):
  - Motor torque: $T_m(t) = k_t I_a(t)$ [Newton-meters]
  - Back EMF voltage: $E_c(t) = k_e \dot{\theta}(t)$ [volts]

7.1 Franklin *et al.* Example 2.14 (7th ed,), Armature-controlled DC motor

Step 1 Setup model

- Draw a schematic diagram

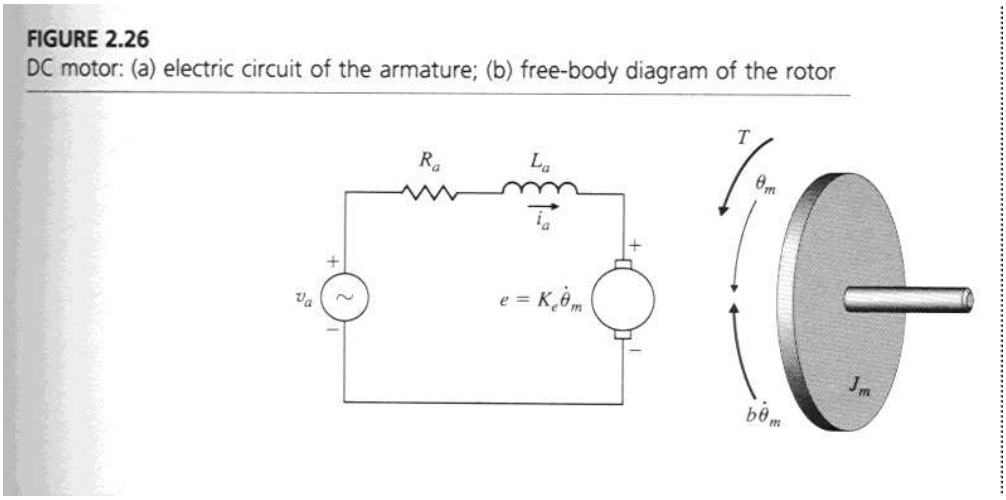


Figure 25: DC motor, electric circuit and mechanical schematic (fig 2.26).

- Assign variables to signals and parameters, see schematic.
- Write the constituent equations for elements of the system
  - Electrical side:

$$\begin{aligned} V_{R_a}(t) &= R_a I_a(t) \\ V_{L_a}(t) &= L_a \frac{d I_a(t)}{dt} \\ E_c(t) &= k_e \dot{\theta}_m(t) \end{aligned}$$

  - Mechanical side:

$$\begin{aligned} M_c \ddot{\theta}(t) &= T_{net.J_m}(t) \\ T_b(t) &= b_c \dot{\theta}(t) \\ T_m(t) &= k_t I_a(t) \end{aligned}$$

Step 2 Write the continuity equations describing the connections of the system

- Electrical side, KVL:

$$V_{R_a}(t) + V_{L_a}(t) + E_c(t) = V_a(t) \tag{100}$$

- Mechanical side, from free-body diagram

$$J_m \ddot{\theta}(t) = T_m(t) - T_b(t) \tag{101}$$

which gives

$$J_m \ddot{\theta}(t) + b\dot{\theta}(t) = T_m(t) \tag{102}$$

- Combine the constituent and continuity equations to derive the governing differential equations.

$$L_a \dot{I}_a(t) + R_a I_a(t) = V_a(t) - k_e \dot{\theta}(t) \tag{103}$$

$$J_m \ddot{\theta}(t) + b\dot{\theta}(t) = k_t I_a(t) \tag{104}$$

Eqns (103) and (104) have exactly the same form as the voice coil motor model.

Step 3 Put the governing differential equation in standard form

- Eqns (103) and (104) are already in standard form.

Step 4 Derive the transfer function (student exercise)

Step 5 Validating model by verifying units

- Electrical side:

$$L_a \dot{I}_a(t) + R_a I_a(t) = V_a(t) - k_e \dot{\theta}(t) \qquad (103, \text{repeated})$$

$$\left[\frac{volt-second}{amp}\right]\left[\frac{amp}{sec}\right] + \left[\frac{volt}{amp}\right][amp] = [volts] - \left[\frac{volt-sec}{radian}\right]\left[\frac{radian}{sec}\right]$$

giving:

$$[volts] + [volts] = [volts] - [volts]$$

- Mechanical side:

$$J_m \ddot{\theta}(t) + b \dot{\theta}(t) = k_t I_a(t) \qquad (104, \text{repeated})$$

$$\left[\frac{N-m-s^2}{radian}\right]\left[\frac{rad}{s^2}\right] + \left[\frac{N-m-s}{radian}\right]\left[\frac{rad}{s}\right] = \left[\frac{N-m}{amp}\right][amps]$$

giving

$$[N-m] + [N-m] = [N-m]$$

7.2 A reduced-order model for a DC motor

- When the motor electrical time constant is much shorter than the mechanical time constant $\left(\frac{L}{R}\,[\text{sec}] << \frac{J}{b}\,[\text{sec}]\right)$
  - Then the motor inductance can be neglected, and the equation for $i_a(t)$ is simplified, giving a simplified expression for motor torque $T_m(t)$

$$i_a(t) \simeq \frac{V_a(t) - E_c(t)}{R_a} \;; \qquad T_m(t) = k_t\, i_a(t) \simeq k_t \frac{V_a(t) - k_e\, \dot{\theta}_m(t)}{R_a} \tag{105}$$

- Plugging Eqn (105) into Eqn (101) gives

$$J_m\, \ddot{\theta}_m(t) + \left(b + \frac{k_t\, k_e}{R_a}\right) \dot{\theta}_m(t) \;=\; \frac{k_t}{R_a} V_a(t) \tag{106}$$

(Eqn (106) is the motor model used in lab.)

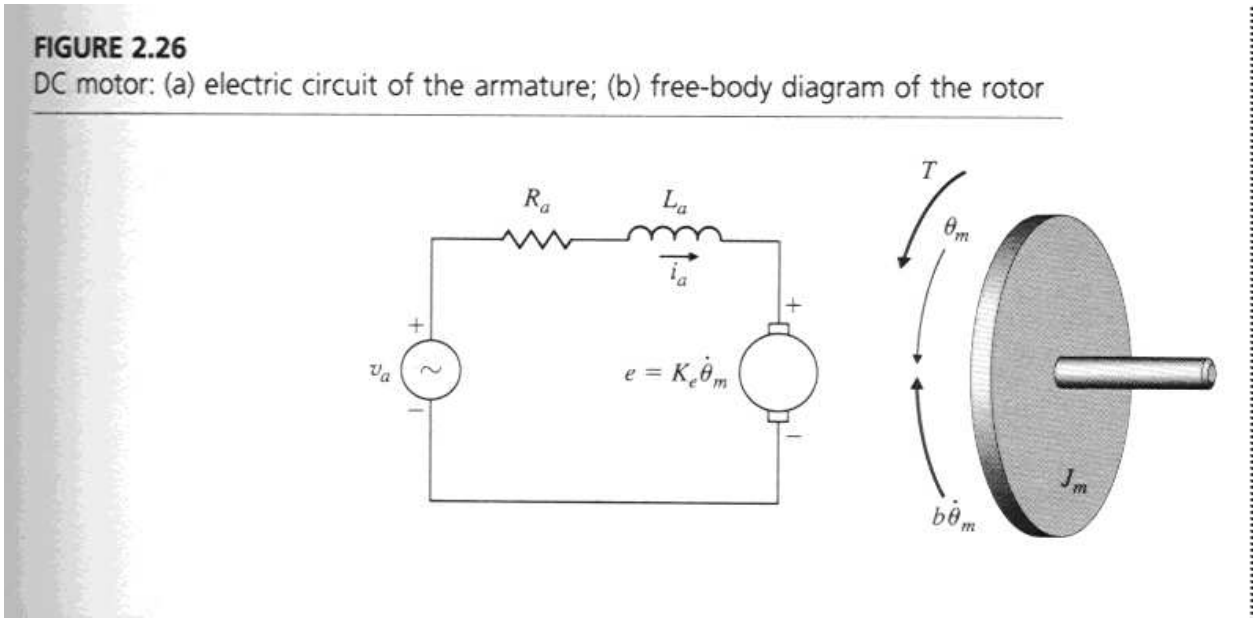


Figure 26: DC motor, electric circuit and mechanical schematic (fig 2.26).

8 AC Motors

- DC Motors require a commutator, which is a point of wear and failure.
- AC motors avoid including a commutation, by implementing a rotating magnetic field.

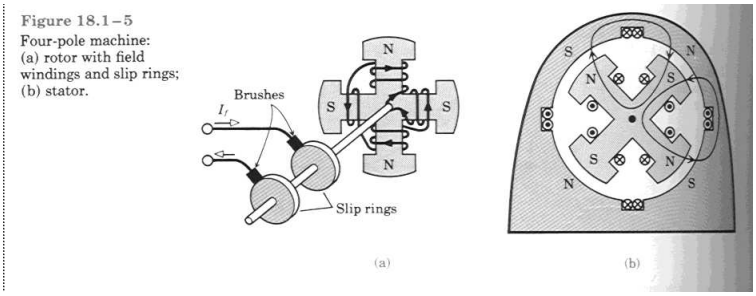


Figure 27: Four-pole AC Motor with slip rings (fig 18.1-5).

Induction Motor:

- No slip rings, rotor windings close a loop.
- Magnetic field rotates at electrical frequency

$$\omega_e(t) = 2\pi 60 \left[\frac{\text{r}}{\text{s}}\right]$$

- Rotor rotates with mechanical rotation rate $\omega_m(t)$, for example

$$\omega_m(t) \simeq 0.98\,\omega_e(t)$$

- The slip of the AC motor is defined as

$$s(t) = \frac{\omega_e(t) - \omega_m(t)}{\omega_e(t)}$$

0.01 to 0.03 is a typical range for slip.

- From the point of view of the rotor, the field is rotating with frequency $\omega_e - \omega_m$, transformer action induces current in the rotor windings.
- Angle γ between rotor current and stator field gives the torque.
- Empirical torque relationship for AC induction motors, based on Fitting a line to the torque speed curve with parameters K_1 and K_2 is:

$$T_m(t) = K_1 V_a(t) - K_2 \dot{\theta}_m(t) \tag{107}$$

- Using parameters K_1 and K_2, we can write the motor dynamics with

$$J_m \ddot{\theta}_m(t) + (b + K_2)\, \dot{\theta}_m(t) \;=\; K_1 V_a(t) \tag{108}$$

  Compare with Eqn (106).

8.1 Conclusion on motors

- There are many different electric machines, both DC and AC. One must look up model details for specific cases.

FIGURE 2.27
Torque-speed curves for two typical induction motors: (a) low-rotor-resistance machine; (b) high-rotor-resistance machine showing four values of armature voltage v_a.

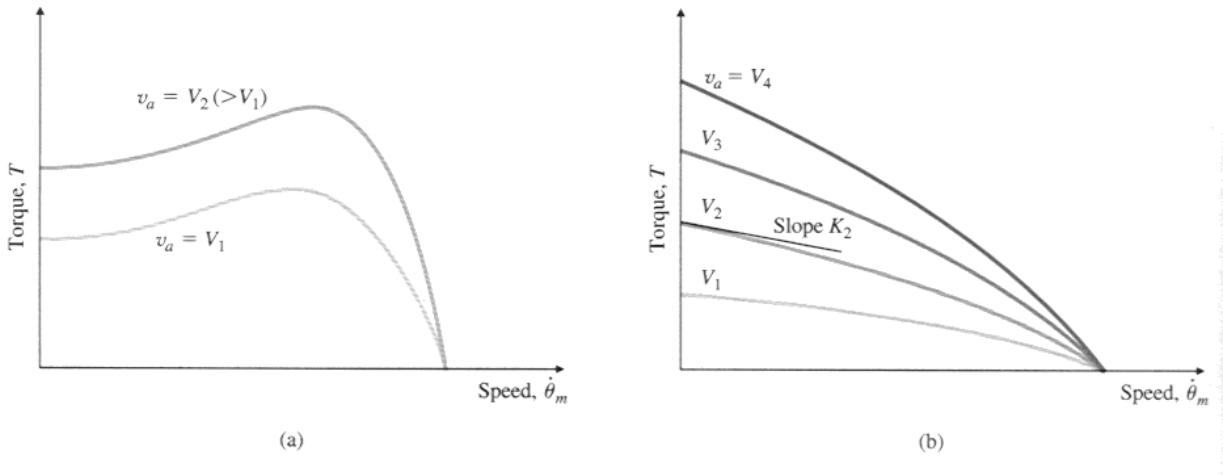


Figure 28: Torque-speed curves (fig 2.27).

9 Dynamics by analogy

Controls engineers are concerned with dynamics in many domains

- Electrical (communications equipment, power control)
- Electro-mechanical (Motor drives, robotics, automation)
- Mechanical (Robotics, factory automation)
- Fluid (Flight controls, oil refinery, process industries)
- Thermal (oil refinery, process industries, energy generation)

At first glance, these different domains seem complicated, they each have their own

- Schematic language (e.g., circuit schematics or free-body diagrams)
- Basic modeling equations (e.g., Ohm's law or Newton's laws)
- Technology

But actually, for dynamic modeling they have more in common than differences.

It is possible to model Mechanical / Fluid / Thermal systems as Electrical Circuits by using ***Through*** and ***Across Variables***.

9.1 Through and across variables

Each domain has a variable representing a flow (through variable) and a variable representing a pressure (across variable)

Through Variable: represents a flow, determined by integrating over an area

- Current (integral of charge flux across conductor area)
- Total Force (integral of stress across a cross section)
- Fluid Flow (integral of flow across pipe area)
- Total Heat Flux (integral of heat flux across heat conducting area)

Across Variable: measured between two points.

- Voltage (measured with respect to ground)
- Velocity or position (measured with respect to a reference frame)
- Pressure (measured with respect to a defined 0 pressure)
- Temperature (measured with respect to a defined 0 temperature)

10 Summary and conclusions

- We've seen an approach to modeling devices based on five steps

  1. Setup model
  2. Write and arrange constituent equations
  3. Write and arrange continuity equations
  4. Write the transfer function
  5. Validate

- We've worked through modeling systems of several types
  - Mechanical
  - Electrical
  - Electronic, with an op amp
  - Electro-mechanical, motors

- We've seen that mechanical, fluid and thermal systems can all be modeled by an electrical analogy, and how to develop the analogous circuit.

- We've seen the basics of forming and working with transfer functions.

- Conclusion

  Modeling is the first step in controller development. The controls engineer should be able to model diverse mechanical, electrical and other systems.